

APPENDIX B

DECLARATION AND POWER OF ATTORNEY

As a below named inventor, I hereby declare that:

My residence, post office address, and citizenship are as stated below next to my name,

I believe I am the original, first and sole inventor (if only one name is listed below) or an original, first and joint inventor (if plural names are listed below) of the subject matter claimed and for which a patent is sought on the invention entitled:

HIGH VOLTAGE DEVICE

the specification of which

☒ is attached hereto

☐ was filed on ___ as Application Serial No. ___ and was amended on ___ (if applicable).

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claims, as amended by any amendment referred to above.

I do not know and do not believe the same was ever known or used in the United States of America before this invention thereof or more than one year prior to this application, and that the same was not in public use or on sale in the United States of America more than one year prior to this application. I acknowledge the duty to disclose information which is known to me to be material to patentability in accordance with title 37, Code of Federal Regulations, section 1.56.

I hereby claim foreign priority benefits under Title 35, United States Code, Section 119(a)-(d) or Section 365(b) of any foreign application(s) for patent or inventor's certificate, or Section 365(a) of any PCT international application which designated at least one country other than the United States, listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed:

Prior Foreign Application(s):

<u>Number</u>	<u>Country</u>	<u>Day/Month/Year filed</u>	<u>Priority Claimed</u>	
			<u>Yes</u>	<u>No</u>
				☒

I hereby claim the benefit under 35 USC 119(e) of any United States provisional application(s) listed below:

Prior Provisional Application(s):

<u>Application Number</u>	<u>Filing Date</u>

I hereby claim the benefit under Title 35, United States Code, Section 120 of any United States application(s), or Section 365(c) of any PCT international application designating the United States, listed below and, insofar as the subject matter of each of the claims of this application is not disclosed in the prior United States or PCT international application in the manner provided by the first paragraph of Title 35, United States Code, Section 112, I acknowledge the duty to disclose material information as defined in Title 37, Code of Federal

Regulations, Section 1.56 which occurred between the filing date of the prior application and the national or PCT international filing date of this application:

Prior U.S. Application(s):

<u>Serial No.</u>	<u>Filing Date</u>	<u>Status: Patented, Pending, Abandoned</u>
12/390,509	02/23/2009	Pending

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I hereby revoke any previous Powers of Attorney and appoint the following attorney(s) and/or agent(s), each said individual being a member or associate of **Horizon IP Pte. Ltd.** and associated with:

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for so long as they remain with such law offices or company with full power of substitution and revocation, to prosecute this application and to transact in connection therewith all business in the Patent and Trademark Office and before competent International Authorities; said appointment to be to the exclusion of myself and my other attorney(s).

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HIGH VOLTAGE DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a divisional application which claims benefits of co-pending U.S. patent application Ser. No. 12/500,620, filed on July 10, 2009, which is a continuation-in-part application of U.S. patent application Ser. No. 12/390,509, filed on February 23, 2009, now issued as U.S. patent No. 8,053,319. All disclosures are incorporated herewith by reference.

BACKGROUND

[0002] Lateral Double-Diffused (LD) transistors have been widely employed in high voltage applications. For high performance LD transistors, low drain-to-source on-resistance ($R_{ds_{on}}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability. To achieve low $R_{ds_{on}}$, the channel of the high LD transistor should be as short as possible.

[0003] As technology evolves into era of deep sub-micron (e.g., beyond 0.35 μm) Very-Large-Scale Integration (VLSI), there is a desire for both high voltage (HV) LD transistors and low voltage (LV) transistors to be fabricated on the same substrate. Generally, the processes for forming LV devices, such as complementary metal-oxide-semiconductor (CMOS) processes, are used to form the HV devices. However, conventional CMOS processes are incompatible for forming HV transistors. For example, process overlay issues make it difficult for alignment of the HV channel well, which defines the channel length. Process variations in process overlay requires large process windows. This makes forming short channel lengths to achieve low $R_{ds_{on}}$ difficult if not impossible. Furthermore, the thin gate electrodes used prevent them serving as a hard mask, further exacerbating process control in forming the channel well.

[0004] From the foregoing discussion, it is desirable to provide reliable HV devices formed with short channel lengths to have low $R_{ds_{on}}$.

SUMMARY

[0005] A device is disclosed. In one embodiment, the device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device further includes a doped drift well adjacent to a second edge of the gate.

[0006] In another embodiment, another device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device also includes a doped drift well adjacent to a second edge of the gate. The device further includes a drift isolation region in the doped drift well.

[0007] In yet another embodiment, a semiconductor device is disclosed. The semiconductor device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device also includes a doped drift well adjacent to a second edge of the gate. A drift

isolation region is disposed in the doped drift well. The device further includes first and second diffusion regions adjacent to the gate.

[0008] These and other objects, along with advantages and features of the present invention herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and can exist in various combinations and permutations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily to scale, emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:

[0010] Fig. 1 shows an embodiment of a device;

[0011] Figs. 2a-g show an embodiment of a process for forming a device;

[0012] Figs. 3a-g show an alternative embodiment of a process for forming a device;

[0013] Figs. 4a-g show another embodiment of a process for forming a device; and

[0014] Figs. 5a-g show another embodiment of a process for forming a device.

DETAILED DESCRIPTION

[0015] Embodiments generally relate to semiconductor devices or integrated circuits. More particularly, some embodiments relate to high power devices. For example, high power devices include lateral double-diffused (LD) transistors, such as metal oxide transistors (MOS). The high power devices can be employed as switching voltage regulators for power management applications. The LD transistors can be easily integrated into devices or ICs. The devices or ICs

can be incorporated into or used with, for example, consumer electronic products, particularly portable consumer products, such as cell phones, laptop computers and personal digital assistants (PDAs).

[0016] Fig. 1 shows a portion 100 of an embodiment of a device. As shown, the portion includes a substrate 101. The substrate, for example, comprises a silicon substrate. The substrate can be lightly doped with p-type dopants. N-type or other types of substrates, such as silicon germanium or silicon-on-insulator (SOI), are also useful.

[0017] The substrate may be prepared with first and second regions 110 and 160. In one embodiment, the first region comprises a low voltage (LV) device region while the second region comprises a high voltage (HV) device region. The regions may be other types of device regions or additional device regions may be provided. For example, a dual gate oxide (DGO) region may be provided for DGO devices. The DGO devices may be used for I/O circuitries. Alternatively, the device may include only a HV region.

[0018] The first region, in one embodiment, comprises first and second type active sub-regions 112 and 116. The first and second types are complementary types, forming a complementary type device. For example, the complementary type device comprises a complementary metal oxide semiconductor (CMOS) device. Forming non-complementary or other types of devices is also useful.

[0019] The first type active sub-region, for example, comprises a n-type active sub-region and the second type active sub-region comprises a p-type active region. The first type active sub-region comprises a second type doped well 122; the second type active sub-region comprises a first type doped well 124. For example, the first type active sub-region comprises a p-type doped well and the second type active sub-region comprises a n-type doped well. Other configurations of active sub-regions are also useful. P-type dopants can include boron (B),

indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof.

[0020] A first type transistor 140 is provided in the first type active sub-region and a second type transistor 150 is provided in the second type active sub-region. For example, a n-type transistor is provided in the n-type active sub-region with a p-type doped well while a p-type transistor is provided in the p-type active sub-region with a n-type doped well. The transistors, in one embodiment, are MOS field effect transistors (MOSFETs). Other types of transistors may also be useful.

[0021] A transistor, for example, comprises a gate having a gate electrode 144 over a gate dielectric 142. The gate electrode, for example, comprises polysilicon. Other types of gate electrode materials, such as polysilicide or metal silicide, are also useful. Polysilicide or metal silicide gates can be formed by reacting the desired metal with polysilicon. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. Doping the gate electrode may depend on the technology. The gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the gate electrode with other dopant types is also useful. The gate dielectric can be silicon oxide. Other types of gate electrode or gate dielectric materials can also be useful. The material of the gate electrode and gate dielectric may depend on the technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

[0022] Sidewalls of the gate can be provided with sidewall spacers 148. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

[0023] Adjacent to the gate are diffusion regions 146. The diffusion regions serve as, for example, source and drain of the transistor. Extension diffusion regions 147 can be provided. The extension diffusion regions, for example, are shallow diffusion regions which extend under the spacers. The diffusion regions (including the extension diffusion regions) of the first type transistor are doped with the first type dopants and the diffusion regions of the second type transistor are doped with the second type dopants. For example p-type diffusion regions are provided for p-type transistors and n-type diffusion regions are provided for n-type transistors. The depth and dopant concentration of the diffusion regions may depend, for example, on the application, such as voltage requirements.

[0024] The HV region includes a HV active region 165. In one embodiment, the HV active region comprises a first type active region. For example, the HV active region comprises a n-type active region. Providing a p-type HV active region is also useful. Furthermore, it is understood that the HV region is depicted with only one active region for purpose of simplification and that the HV region can include numerous HV active regions or sub-regions. For example, the HV active sub-regions can be complementary types of HV sub-regions for complementary types of HV devices.

[0025] Isolation regions 180, such as shallow trench isolation (STI) regions are provided to isolate the LV and HV regions as well as active sub-regions within the regions. Other types of isolation regions are also useful.

[0026] In one embodiment, the first type HV active region comprises at least first and second portions. The first portion comprises a second type HV channel (or body) well 135 and the second portion comprises a first type HV drift well 136. The HV drift well, for example, defines the type of HV device. For example, a p-type HV active region would have a n-type HV channel well and a p-type HV drift well. In one embodiment, a third or middle portion separates the

channel and drift wells. In other embodiments, the first and second portions are contiguous portions. For example, the HV channel and HV drift wells contact each other.

[0027] A deep well 132 may be provided in the HV active region, encompassing the HV channel and drift wells. Generally, the deep well comprises dopants of the first type or same as the HV device type. The deep well may be used to isolate the drift well from the substrate. For example, a p-type device will have a n-type deep well. In some cases, the deep well can be optional or the deep well can comprise dopants of the second or opposite type of the HV device type. For example, in the case of a n-type HV device, the deep well can be a p-type or n-type deep well. The different types of deep well selected may depend on the type of substrate. Generally, p-type doped substrates are used for fabricating devices. When p-type substrates are used, the n-type deep well may be optional. This is because the n-type drift well is already different from the p-type substrate. For a n-type substrate, a p-type deep well may be used to isolate the channel well from the substrate. Other configurations of deep wells may also be useful.

[0028] As shown, a first type HV device is provided in the first type HV active region. The first type HV device, in one embodiment, comprises a LD device, such as a LDMOS device. The HV device comprises a gate with first and second edges 179a-b or sidewalls.

[0029] The gate includes a HV gate electrode 174 over a HV gate dielectric 172. The HV gate electrode, for example, comprises polysilicon. Other types of HV gate electrode materials, such as polysilicide or metal silicide, are also useful. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of HV gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. The HV gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the HV gate electrode with other dopant types is also useful. The HV gate dielectric can be silicon oxide. Other types of HV gate dielectric materials can also be useful.

[0030] In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process and/or technology. In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process for LV devices and/or technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

[0031] Sidewalls of the gate can be provided with sidewall spacers 178. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

[0032] In one embodiment, the various gate elements of the HV gate, such as gate electrode, gate dielectric and spacers are of the same type of material as the LV devices in the LV region. This facilitates process compatibility for forming HV and LV devices.

[0033] The HV gate overlaps the channel and drift wells in the first and second portions of the HV active region. A drift isolation region 180a may be provided within the drift well 136 on the high voltage side of the HV device. The drift isolation region 180a can improve breakdown voltage of the HV device due to high voltage. For example, the drift isolation region is particularly useful to sustain high voltages for applications beyond about 10V. As shown, the drift isolation region 180a is disposed in the drift well 136 under the gate. Disposing the drift well at other positions, such as partially underlying the gate may also be useful.

[0034] A first type drain diffusion region 176b is provided between the drift isolation region 180a and isolation region 180 which isolates the HV active region from other sub-active regions in the HV region. A first type source diffusion region 176a is provided in the channel well 135 of the first type HV active region. A source extension region 177 can be provided which extends under the spacer on the channel side of the HV gate. The first type source and drain diffusion regions, for example, comprise a p-type diffusion regions for a p-type HV active region.

Providing n-type diffusion regions for a n-type HV active region is also useful. The dopant depth and concentration of the diffusion regions may be selected depending on the application. In one embodiment, the dopant depth and concentration of diffusion regions may be the same as those of LV devices. Other dopant depths and concentrations are also useful.

[0035] A second type body contact region 171 may be provided in the first portion of the HV active region. As shown, the second type body contact region 171 is between the source diffusion region 176a and the isolation region 180. The second type body contact region comprises second type dopants. In one embodiment, dopant depth and concentration of the body contact region may be the same as the diffusion regions of the LV devices. Other dopant depths and concentrations are also useful. The body contact region 171 provides electrical coupling to the channel well 135.

[0036] An edge 133 of the channel well 135 extends below the gate on the source side of the gate. The edge 133, for example, can be referred to as the channel edge of the channel well. The channel edge of the channel well 135 and the first edge 179a of the gate on the source side define a channel of the HV transistor having an effective channel length L.

[0037] The effective channel length, in one embodiment, is controlled by ion implantation which is self-aligned to the gate edge. For example, dopant ions are implanted using an implant mask through the gate electrode or a hard mask and the gate electrode. In one embodiment, the implant mask is also used to pattern the channel edge of the HV gate. The effective channel length may be controlled by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements. For example, shorter effective channel length may be achieved by decreasing tilt angle, implant energy or dosage and vice-versa.

[0038] With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided. Very short effective channel lengths with very low $R_{ds(on)}$ can be achieved, enabling high current to pass through a small area. This is particularly useful for

technologies beyond 0.35 μm , such as, for example, 0.25 μm , 0.18 μm and 0.13 μm technologies.

[0039] Smaller area requirements for power management applications with reduced power dissipation increases performance while reducing costs.

[0040] In one embodiment, the effective channel length L is less than 0.4 μm . In other embodiments, the effective channel length is less than 0.3 μm . In yet another embodiment, the effective channel length is less than 0.25 μm . Other effective channel lengths may also be useful, for example, depending on the application.

[0041] Figs. 2a-g show cross-sectional views of an embodiment of a process for forming a device or IC 100. Referring to Fig. 2a, a substrate 101 is provided. The substrate can comprise a silicon substrate, such as a lightly p-type doped substrate. Other types of substrates, including n-type doped substrates, silicon germanium or silicon-on-insulator (SOI), are also useful.

[0042] As shown, first and second device regions 110 and 160 are defined on the substrate. The first region, for example, comprises a LV device region while the second region comprises a HV device region. Other types of device regions or additional device regions may be provided. Alternatively, the device may include only a HV region.

[0043] The LV device region 110 can be defined with first and second type active sub-regions 112 and 116. In one embodiment, the first and second type active sub-regions are complementary type active sub-regions to form a complementary device such as a CMOS device. Forming non-complementary or other types of devices is also useful.

[0044] As for the HV device region 160, a HV active region 165 is defined therein. In one embodiment, the HV active region 165 can be a first or second type HV active region. Although only one HV active region is depicted, it is understood that the HV device region may include additional HV active regions or sub-regions. The HV active sub-regions can be complementary HV active sub-regions.

[0045] The substrate is also prepared with isolation regions 180 to separate the device regions as well as other active device sub-regions. In one embodiment, the isolation regions 180 comprise STIs. Various conventional processes can be employed to form the STI regions. For example, the substrate can be etched using conventional etch and mask techniques to form trenches which are then filled with dielectric materials such as silicon oxide. Chemical mechanical polishing (CMP) can be performed to remove the excess dielectric materials and provide a planar substrate top surface. Other processes or materials can also be used to form the STIs.

[0046] In Fig. 2b, the doped wells for the active regions of the device regions are formed. In one embodiment, a first type doped well 124 is formed in the second type LV active sub-region 116 and a second type doped well 122 is formed in the first type LV active sub-region 112. For example, a n-type doped well is formed in the p-type active sub-region and a p-type doped well is formed in the n-type active sub-region.

[0047] In the HV active region 165, it includes at least first and second portions. For example, the first portion corresponds to a drain portion and the second portion corresponds to a source portion. In some embodiments, a third or middle portion is provided to separate the first and second portions. Alternatively, the first and second portions are contiguous portions without a middle portion.

[0048] In one embodiment, a doped well 136 is formed in the drain portion of the HV active region. The doped well, for example, is a drift well of the HV device. The drift well comprises the same type as the HV active region. In the case of a first type HV active region, the drift well comprises a first type. For example, a n-type drift well is provided for a n-type HV active region.

[0049] A drift STI region 180a may be optionally provided in the drift well 136. The drift STI region is disposed in the drift well to improve reliability of high voltage devices. For

example, the drift STI region improves breakdown reliability of the HV device. The drift STI region can be formed at the same time as the other STI regions.

[0050] A deep HV well 132 may be provided in the HV active region 165. As shown, the deep HV well 132 is formed in the whole HV active region 165 and encompasses the first and second portions of the HV active region 165. The deep HV well comprises an opposite type as the drift well and the HV active region. In the case of a first type HV active region, the deep well comprises a second type. For example, a p-type deep well is provided for a n-type HV active region.

[0051] The depth and concentration of dopants of the various wells may depend on, for example, application. For example, higher voltage applications may require lower doping concentration of dopants and greater depth of the wells. In one embodiment, the depth of the deep well 132 is about 3 μm with a dopant concentration of about $5 \times 10^{16} \text{cm}^{-3}$. For the drift well 136, it may have a depth of about 1.5 μm with a dopant concentration of about $5 \times 10^{17} \text{cm}^{-3}$. Other depths and dopant concentrations are also useful for the wells.

[0052] As described, the STI regions are formed prior to forming the various doped wells. Other process schemes, such as forming the STI regions after the various doped wells are formed may also be useful.

[0053] The doped wells can be formed by ion implantation. P-type dopants can include boron (B), BF_2 (boron compounded with fluorine), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof. Generally, the first and second types of the active regions are formed selectively in separate processes. For example, the n-type regions can be implanted with n-type dopants while an implant mask prevents doping of the p-type regions.

[0054] In one embodiment, the wells of the HV region are formed separately from the LV regions. For example, the LV wells can be formed first followed by the HV wells. Forming the

HV wells prior to the LV wells is also useful. The different types of LV wells can be formed in separate processes and the different types of HV wells can also be formed in separate processes. For example, the first type and second type wells in the LV region are formed in separate implant processes. Similarly, the different types of wells in the HV region are formed in separate implant processes.

[0055] A doped well can be formed by a single implant process or multiple implant processes. In multiple implant processes, dopants can be implanted at different energy levels to achieve the desired doping profiles. In the case of a single implant process, a high temperature drive-in process is performed to achieve the desired doping profile.

[0056] A single implant process may be employed to, for example, form deep wells. For a n-type deep well, phosphorus may be implanted with a dopant concentration of about $6 \times 10^{12} \text{ cm}^{-2}$ at an energy level of about 2000 KeV followed by a two-hour drive-in at a temperature of about 1100 °C. Single implant processes may also be used to form other types of wells. Forming wells using multiple implant processes may also be useful. For example, LV wells may be formed by multiple implant processes.

[0057] In Fig. 2c, various layers of the gate are formed on the substrate. In one embodiment, a gate dielectric layer 242 is formed over the substrate. The gate dielectric layer, for example, comprises silicon oxide. Other types of dielectric materials, such as silicon oxynitride, may also be useful. Alternatively, high-k, low-k or composite of dielectric materials may be used. The thickness of the gate dielectric layer 242 may be about 30 Å for 1.8V gate voltage transistors. Other thicknesses may also be useful. The thickness, for example, may depend on the gate voltage applications. For example, higher gate voltages may require thicker gate dielectric layers. In one embodiment, the gate dielectric layer is formed by thermal oxidation. Other techniques, such as chemical vapor deposition (CVD), can also be used to form the gate dielectric layer.

[0058] A gate electrode layer 244 is deposited on the gate dielectric layer 242. The gate electrode layer comprises, in one embodiment, polysilicon (poly). The gate electrode layer 244 can be formed as an amorphous or non-amorphous layer. For amorphous deposited layer, subsequent processing can be performed to crystallize it. Other types of gate electrode materials are also useful. For example, the polysilicon can be subsequently processed to form polysilicide or metal gates. The thickness of the gate electrode layer may be about 2000 Å or less. In another embodiment, the thickness of the gate electrode layer is about 2500 Å or less. In yet another embodiment, the thickness of the gate electrode layer can be about 4000 Å or less. For example, in the case of polysilicide gates, the polysilicon may be about 2000 Å while the tungsten is about 2000 Å. Other thicknesses are also useful. Various techniques can be used to form the gate electrode layer. For example, polysilicon can be deposited by CVD while metal can be deposited by sputtering. Other techniques, depending on the material, may also be useful.

[0059] Referring to Fig. 2d, a mask layer 288 is formed on the substrate, covering the gate electrode layer. In one embodiment, the mask layer comprises photoresist. The mask is patterned as desired. To pattern the mask layer, photolithography can be employed. For example, the mask can be selectively exposed to an exposure source through a lithographic mask. Depending on whether a positive or negative resist is used, the exposed or unexposed portions are removed by development. To enhance lithographic resolution, an ARC layer (not shown) may be provided beneath the mask layer.

[0060] The mask layer 288 is patterned to form an opening 289 to expose a portion of the gate electrode layer 244. In one embodiment, the opening corresponds to the opening for the channel well implant. For example, the mask layer serves as the channel well implant mask. The ARC layer may be patterned with the mask layer. Subsequently patterning the ARC layer may also be useful.

[0061] In Fig. 2e, the substrate is implanted with dopants to form a channel well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region. In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0062] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0063] After forming the channel doped well 135, exposed portions of the gate electrode layer 244 are removed, as shown in Fig. 2f. In one embodiment, an anisotropic etch, such as a reactive ion etch (RIE) is performed to remove exposed portions of the gate electrode layer. The removal of the exposed portions of the gate electrode layer, in one embodiment, forms a first edge of a HV gate. The first edge, in one embodiment, corresponds to the source side of the HV gate. As shown, the gate dielectric layer 242 serves as an etch stop for etching the gate electrode layer. Leaving the gate dielectric layer can be advantageous since it can serve as an implant mask for subsequent ion implantation to protect the substrate from implant damage.

Alternatively, the exposed portion of the gate dielectric layer can be removed. Removing the gate dielectric layer 242 to expose the substrate may also be useful. After etching the gate electrode layer, the mask layer 288 is removed.

[0064] As described, the implant mask, in one embodiment, serves as a channel well implant mask as well as a mask for patterning the first edge (e.g., source side) of the HV gate. By using the same mask to form the channel well and source side of the gate, the channel is self-aligned to the gate edge. This enables the channel length of the HV device to be well controlled to produce very short effective channel lengths to reduce $R_{ds_{on}}$ performance. In one embodiment, the effective channel length L is less than $0.4\ \mu\text{m}$. In other embodiments, the effective channel length is less than $0.3\ \mu\text{m}$. In yet another embodiment, the effective channel length is less than $0.25\ \mu\text{m}$.

[0065] Referring to Fig. 2g, another mask layer 288b is formed on the substrate and patterned. The mask layer is patterned to expose portions of the gate electrode layer 244 to be removed, protecting portions corresponding to gates of the LV devices. For example, the mask serves as a gate mask. Additionally, the mask layer 288b protects the source portion of the HV region and the gate of the HV device. The exposed portions of the gate electrode layer 244 are removed to form gates of both the LV and HV devices.

[0066] After the gates are formed, the process continues to form the device, as illustrated in Fig. 1. The process, for example, includes forming extension regions 147 by ion implantation, followed by forming spacers 148 and 178 and diffusion regions 146 and 176. The different types of extension and diffusion regions can be formed in different processes. For example, n-type diffusion regions are formed in one implant process and p-type diffusion regions are formed in another process. Furthermore, the diffusion and extension regions of the different device regions can be formed separately. Silicide gates and contacts can be formed after the diffusion regions are formed. For example, metals, such as cobalt are deposited on the substrate 101 and reacted

to form silicide contacts and gates. The contacts and gates can be formed in the same or different processes. Unreacted metals are removed. A pre-metal and an inter-level dielectric layer can be formed in which contacts and interconnects are formed. Additional interconnect levels can be formed by, for example, dual damascene techniques. Final passivation, dicing, assembly and testing may be performed to complete the IC.

[0067] An alternative embodiment of a process of forming a device 100 is shown in Figs. 3a-g. Referring to Fig. 3a, a partially processed device is shown. The partially processed device is similar to that shown in Fig. 2c. A hard mask 388 is formed on the gate electrode layer 244. The hard mask, for example, comprises silicon oxide. In one embodiment, the hard mask 388 is formed by CVD. The thickness of the hard mask, for example, is about 40nm. Other materials, techniques or thicknesses may also be useful. For example, the hard mask may be formed from silicon nitride or other types of materials.

[0068] Referring to Fig. 3b, a soft mask 287, such as photoresist, is formed on the hard mask 388. The soft mask can be patterned by photolithographic techniques to form an opening 289 which can serve as a channel implant mask. An ARC layer can be provided between the soft mask 287 and the hard mask 388. The pattern of the soft mask is transferred to the hard mask by, for example, RIE.

[0069] In one embodiment, the soft mask layer is removed after patterning of the hard mask, as shown in Fig. 3c. The substrate is implanted with dopants to form a channel doped well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

[0070] In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending

on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0071] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0072] As shown in Fig. 3d, another soft mask layer 287 is deposited on the substrate, covering the hard mask layer and exposed gate electrode layer. The soft mask is patterned by lithography. The soft mask protects portions of the hard mask layer 388 corresponding to the transistor gates. A new ARC layer can be provided between the soft mask 287 and the hard mask 388. In Fig. 3e, exposed portions of the hard mask are removed, exposing portions of the gate electrode layer. Patterning of the hard mask layer can be achieved by, for example, RIE. The patterned hard mask serves as a gate mask. The soft mask, as shown in Fig. 3f, is removed after patterning the hard mask.

[0073] Referring to Fig. 3g, the gate electrode layer 244 is patterned to form gates in the LV and HV region by, for example etch. Patterning of the gate electrode layer can be achieved using, for example, RIE. The patterning of the gate electrode layer also erodes the hard mask, reducing its thickness. The hard mask 388 can be removed by, for example, a clean step after etching the gate electrode layer. The process continues as previously described to complete the device.

[0074] Another embodiment of a process for forming a device 100 is shown in Figs. 4a-g. Referring to Fig. 4a, a partially processed device is shown. The partially processed device is similar to that shown in Fig. 3a.

[0075] Referring to Fig. 4b, a soft mask layer 387, such as a photoresist layer, is formed on the hard mask 388. An ARC layer can be provided between the soft mask and the hard mask. The soft mask layer is patterned to expose portions of the hard mask 388. The remaining portion of the soft mask corresponds to the HV gate in the HV region 160. Patterning of the soft mask layer can be achieved by photolithographic techniques. In one embodiment, the patterned soft mask layer also exposes the hard mask in the LV region 110.

[0076] As shown in Fig. 4c, the pattern of the soft mask is transferred to the hard mask by, for example, RIE. The RIE removes exposed portions of the hard mask to expose portions of the gate electrode layer 344 in the HV region which are to be removed. In one embodiment, the removal of the hard mask also exposes the gate electrode layer in the LV region. After patterning of the hard mask, the soft mask layer is removed.

[0077] In Fig. 4d, another soft mask layer 487, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose portions of the gate electrode layer 344 in the LV region which are to be removed, protecting portions corresponding to gates of the LV devices. For example, the patterned soft mask serves as a gate mask for the LV region. The soft mask layer is removed from the HV region.

[0078] The gate electrode layer is patterned in Fig. 4e. In one embodiment, the gate electrode layer in the LV region is patterned using the soft mask while the hard mask is used in the HV region. The use of the soft mask advantageously facilitates forming gates of critical dimensions in the LV region.

[0079] Referring to Fig. 4f, another soft mask layer 487b, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask. The soft mask layer can be patterned by photolithographic techniques to form an opening 489. The patterned soft mask layer serves as a channel implant mask.

[0080] As shown in Fig. 4g, the substrate is implanted with dopants to form a channel doped well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

[0081] In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0082] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0083] In one embodiment, the implant mask and hard mask are removed after forming the channel well. For example, the implant mask is first removed followed by the removal of the

hard mask. In some embodiments, the hard mask is remained. The process continues as previously described to complete the device after the formation of the gates.

[0084] An alternative embodiment of a process for forming a device 100 is shown in Figs. 5a-g. Referring to Fig. 5a, a partially processed device is shown. The partially processed device is similar to that shown in Fig. 4c.

[0085] As shown in Fig. 5b, a soft mask layer 587, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the gate soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose the HV region, protecting the LV devices from being processed. As illustrated by Fig. 5c, the gate electrode in the HV region is patterned by, for example, RIE, to form the gate using the hard mask as the gate mask. After patterning of the gate electrode in the HV region, the soft mask layer is removed.

[0086] Referring to Fig. 5d, another soft mask layer 587b, such as photoresist, is formed over the substrate. An ARC may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to form an opening 589. The patterned soft mask layer serves as a channel implant mask.

[0087] As shown in Fig. 5e, the substrate is implanted with dopants to form a channel doped well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

[0088] In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0089] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0090] Referring to Fig. 5f, another layer of soft mask layer 587c, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer is patterned by photolithographic techniques to expose portions of the gate electrode layer 344 in the LV region to be removed. Additionally, the soft mask layer remains in the HV region to protect it from being processed. The exposed portions of the gate electrode layer 344 are removed to form the gates of the LV device, as illustrated in Fig. 5g.

[0091] Similar to the embodiment of Figs. 4a-g, the LV gates are patterned using a soft mask while a hard mask is used to pattern the HV gate. After forming the LV gates, the soft mask is removed followed by removal of the hard mask over the HV gate. In some embodiments, the hard mask can remain. The process continues to complete the device as described previously.

[0092] The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments, therefore, are to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims, rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

[0093] What is claimed is:

CLAIMS

1. A device comprising:
 - s substrate prepared with an active device region, wherein the active device region includes a gate;
 - a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge defines an effective channel length of the device, wherein the effective channel length is self-aligned to the gate; and
 - a doped drift well adjacent to a second edge of the gate.
2. The device of claim 1 wherein the effective channel length is less than about 0.4 μm .
3. The device of claim 1 wherein the substrate comprises:
 - a drift isolation region in the doped drift well; and
 - a doped deep well in the substrate of the active device region encompassing the doped drift well.
4. The device of claim 1 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.
5. The device of claim 3 wherein the drift isolation region is disposed under the gate.
6. The device of claim 5 wherein first and second diffusion regions are provided adjacent to the gate, wherein:

the first diffusion region is disposed in the doped channel well on a source side of the device; and

the second diffusion region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

7. The device of claim 1 wherein the effective channel length is controlled by ion implantation.

8. The device of claim 7 wherein the ion implantation comprises a tilt angle implant.

9. The device of claim 8 wherein a tilt angle of the ion implantation is about 1-45°.

10. The device of claim 7 wherein the ion implantation comprises multiple tilt angle implants which are rotated about a plane of the substrate.

11. The device of claim 10 wherein a tilt angle of the implant is about 1-45°.

12. The device of claim 7 wherein the ion implantation comprises a quad tilt angle implant which is rotated about a plane of the substrate.

13. A device comprising:

s substrate prepared with an active device region, wherein the active device region includes a gate;

a doped channel well disposed in the substrate adjacent to a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel

well beneath the gate, the first edge of the gate and channel edge defines an effective channel length of the device, wherein the effective channel length is self-aligned to the gate;

a doped drift well adjacent to a second edge of the gate; and

a drift isolation region in the doped drift well.

14. The device of claim 13 wherein the effective channel length is less than about 0.4 μm .

15. The device of claim 13 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

16. The device of claim 1 wherein the drift isolation region is disposed under the gate.

17. The device of claim 13 wherein first and second diffusion regions are provided adjacent to the gate, wherein:

the first diffusion region is disposed in the doped channel well on a source side of the device; and

the second diffusion region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

18. The device of claim 13 wherein the effective channel length is controlled by ion implantation.

19. The device of claim 18 wherein the ion implantation comprises a tilt angle implant, multiple tilt angle implants or quad tilt angle implant.

20. A semiconductor device comprising:

s substrate prepared with an active device region, wherein the active device region includes a gate;

a doped channel well disposed in the substrate adjacent to a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge defines an effective channel length of the device, wherein the effective channel length is self-aligned to the gate; and

a doped drift well adjacent to a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well; and

first and second diffusion regions adjacent to the gate.

HIGH VOLTAGE DEVICE

ABSTRACT

A device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. A doped drift well adjacent to a second edge of the gate is also included.

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

HIGH VOLTAGE DEVICE

First Named Inventor/Applicant Name:

Guowei ZHANG

Filer:

Dexter Kam Chin

Attorney Docket Number:

CSM P 2009 NAT 03 US 2

Filed as Large Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	380	380
Utility Search Fee	1111	1	620	620
Utility Examination Fee	1311	1	250	250

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1250

Electronic Acknowledgement Receipt

EFS ID:	13266067
Application Number:	13550571
International Application Number:	
Confirmation Number:	1089
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Guowei ZHANG
Customer Number:	16162
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 2
Receipt Date:	16-JUL-2012
Filing Date:	
Time Stamp:	23:06:27
Application Type:	Utility under 35 USC 111(a)

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Payment Type	Deposit Account
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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	CSMP2009NAT03US2-ADS.pdf	1096781	no	5
			f4bb40ce5e338b5dfd9d70ec02c39eea05e5d472		
Warnings:					
Information:					
2	Drawings-only black and white line drawings	CSMP2009NAT03US2-Drawings.pdf	203252	no	29
			e76db31f25211664ed6945fe47b81a8770404b16		
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3	Information Disclosure Statement (IDS) Form (SB08)	CSMP2009NAT03US2-IDS.pdf	791077	no	4
			b1de0350bd9ed1dd6cf9229eec66114d1db63a29		
Warnings:					
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4	Oath or Declaration filed	CSMP2009NAT03US2-oath.pdf	84891	no	3
			e30e13b6a08e1268242583be6da53680ff4bead0		
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5	Power of Attorney	CSMP2009NAT03US2-POA.pdf	84891	no	3
			e30e13b6a08e1268242583be6da53680ff4bead0		
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6	Non Patent Literature	CSMP2009NAT03US2-LeonardRubin.pdf	259743	no	8
			a1c44e3b34cf4c7efb5f4949babb59a929451db2		
Warnings:					
Information:					
7		CSMP2009NAT03US2-Specification.pdf	177273	yes	28
			2d1353aa40d29cc3dd8441ba287ccdade262376b		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	23	
	Claims		24	27	
	Abstract		28	28	

Warnings:					
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8	Fee Worksheet (SB06)	fee-info.pdf	32920	no	2
			0934a561467443d59984e60c96e605a24efb67ae		
Warnings:					
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Total Files Size (in bytes):			2730828		
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 2
		Application Number	
Title of Invention	HIGH VOLTAGE DEVICE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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Applicant 1					Remove
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name		Suffix
	Guowei		ZHANG		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Singapore		Country Of Residenceⁱ	SG	
Citizenship under 37 CFR 1.41(b) ⁱ		SG			
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Applicant 2					Remove
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	☐ Party of Interest under 35 U.S.C. 118
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	Purakh Raj		VERMA		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
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Citizenship under 37 CFR 1.41(b) ⁱ		SG			
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All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 2	
		Application Number		
Title of Invention	HIGH VOLTAGE DEVICE			
Customer Number	16162			
Email Address	oa.notification@horizonip.com.sg		Add Email	Remove Email

Application Information:

Title of the Invention	HIGH VOLTAGE DEVICE			
Attorney Docket Number	CSM P 2009 NAT 03 US 2	Small Entity Status Claimed ☐		
Application Type	Nonprovisional			
Subject Matter	Utility			
Suggested Class (if any)		Sub Class (if any)		
Suggested Technology Center (if any)				
Total Number of Drawing Sheets (if any)	29	Suggested Figure for Publication (if any)		

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

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Customer Number	16162		

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This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.			
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Division of	12500620	2009-07-10
Prior Application Status	Patented	Remove	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number		CSM P 2009 NAT 03 US 2	
		Application Number			
Title of Invention		HIGH VOLTAGE DEVICE			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
12500620	Continuation in part of	12390509	2009-02-23	8053319	2011-11-08
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This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).			
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Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
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Assignee 1			Remove
If the Assignee is an Organization check here. ☒			
Organization Name	GLOBALFOUNDRIES Singapore Pte. Ltd.		
Mailing Address Information:			
Address 1	60 Woodlands Industrial Park D Street 2		
Address 2			
City	Singapore	State/Province	
Country ⁱ	SG	Postal Code	738406
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Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature	/dexter chin/			Date (YYYY-MM-DD)	2012-07-16
First Name	Dexter	Last Name	CHIN	Registration Number	38842

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 2
		Application Number	
Title of Invention	HIGH VOLTAGE DEVICE		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

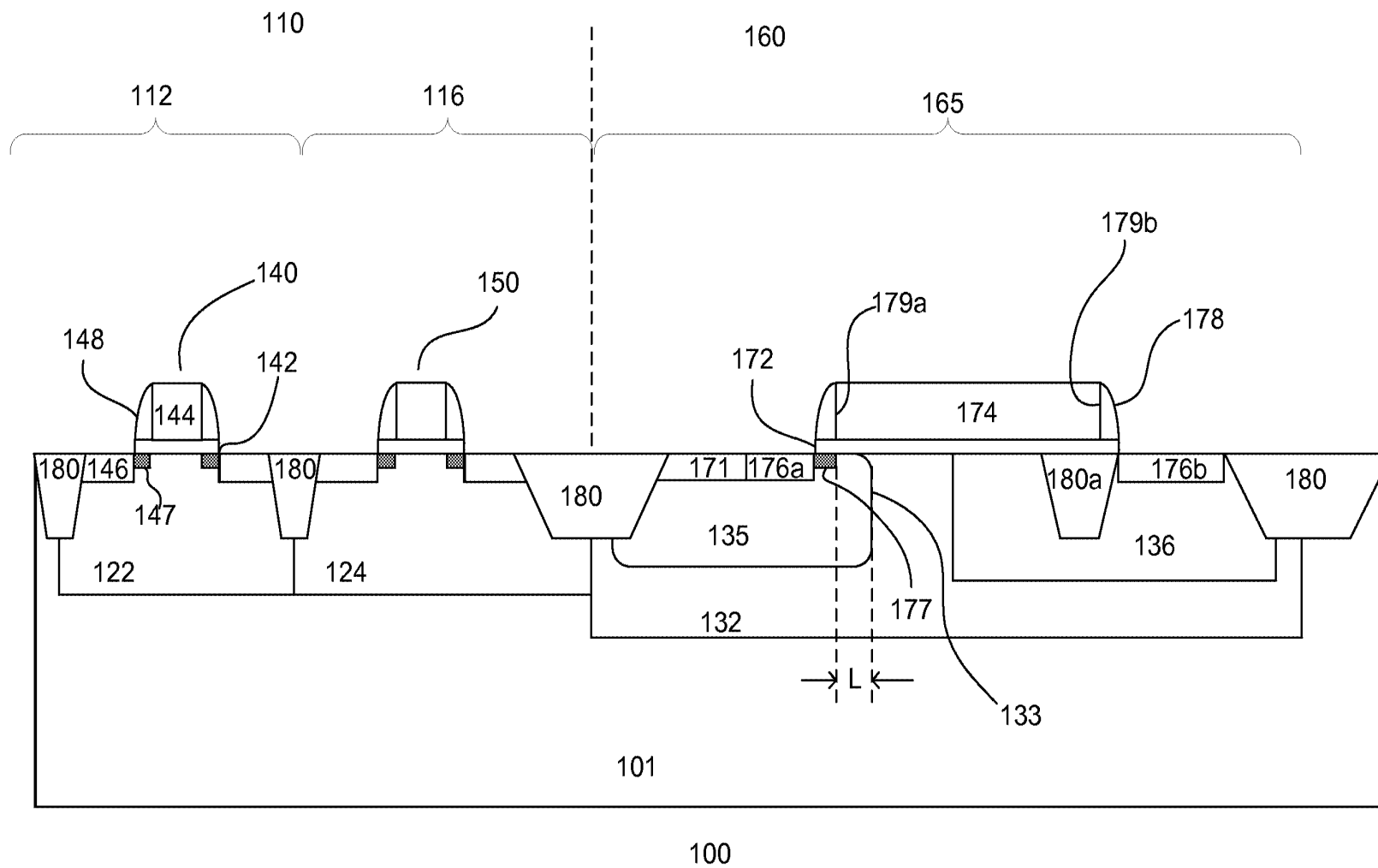


Fig. 1

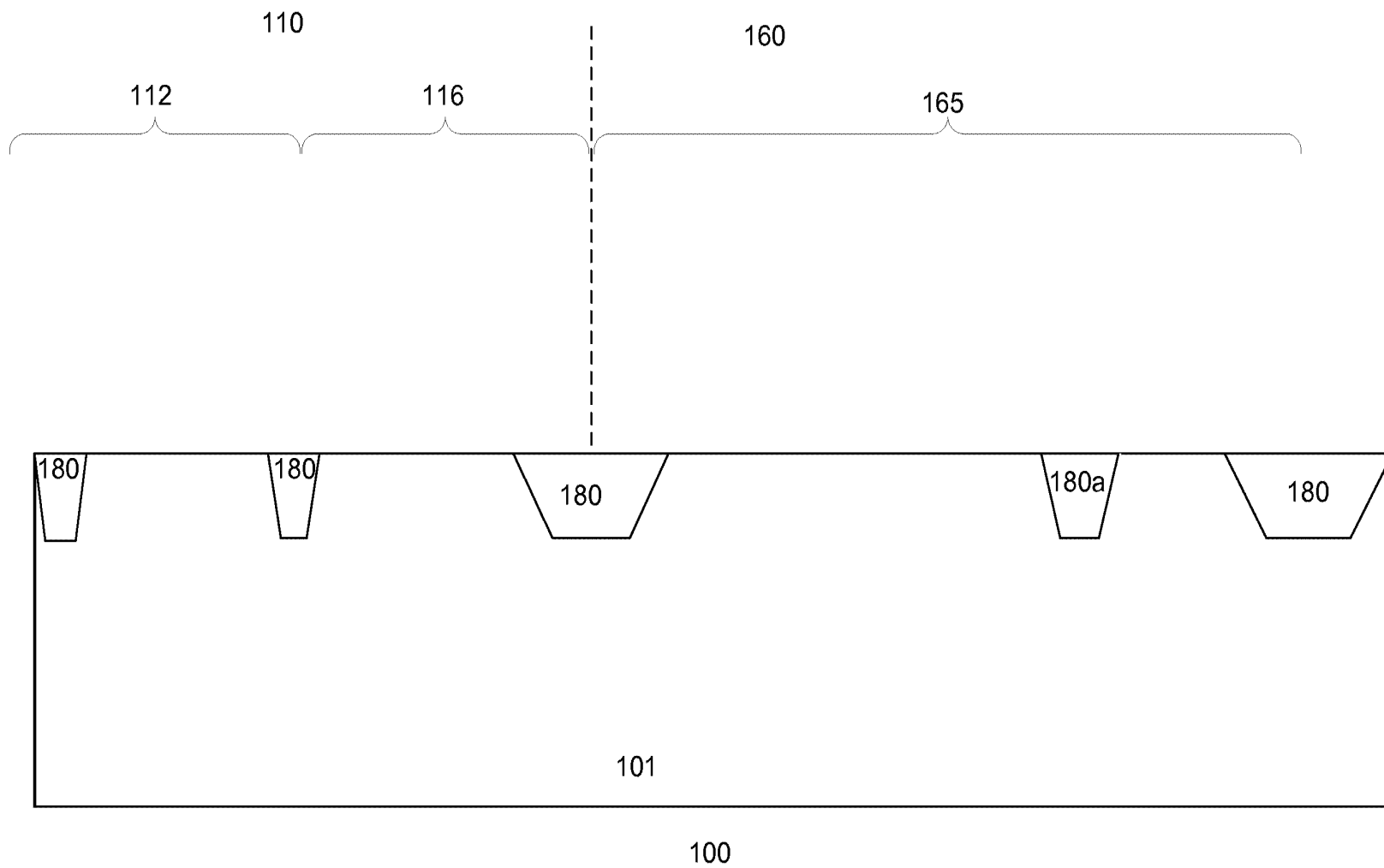


Fig. 2a

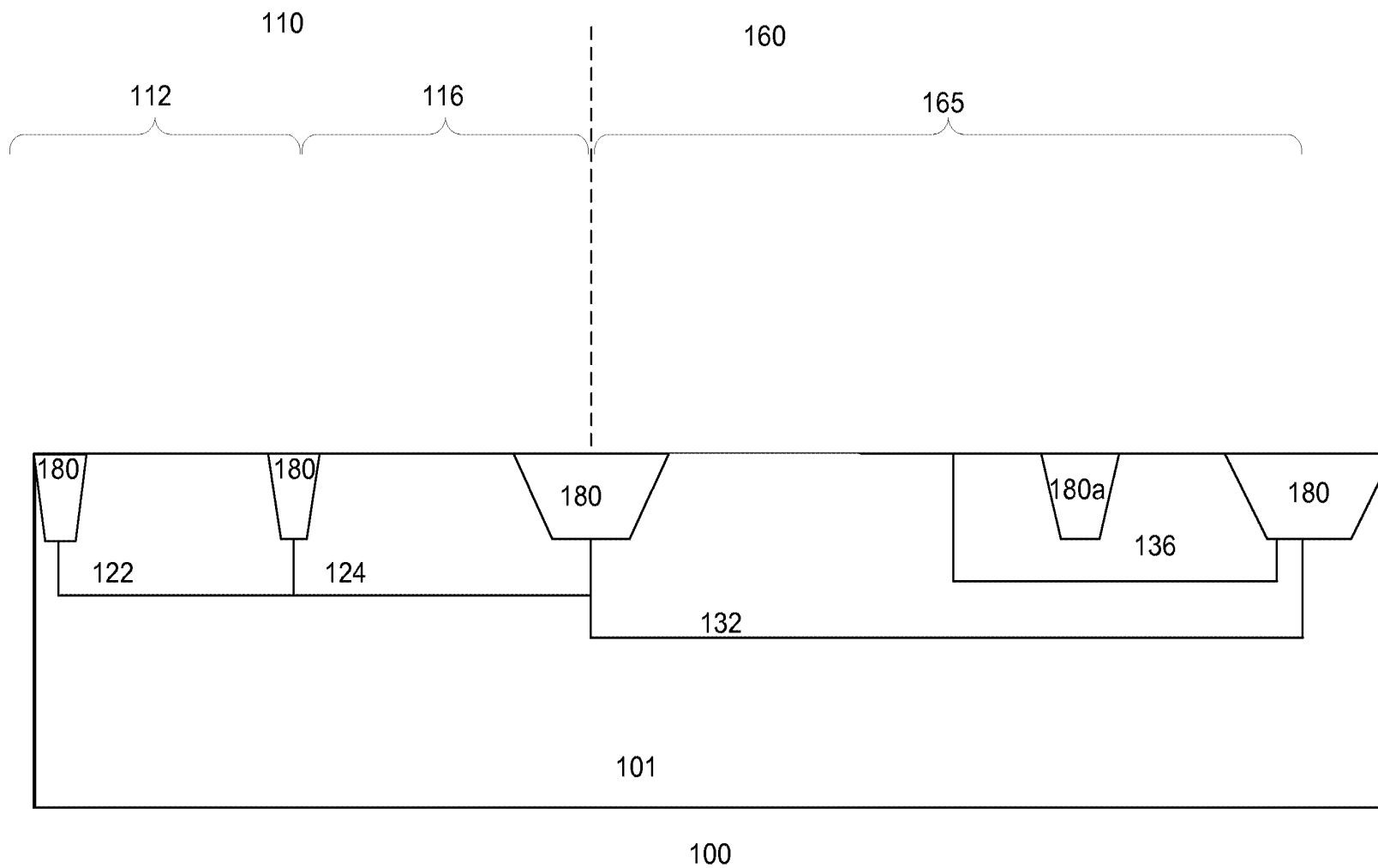


Fig. 2b

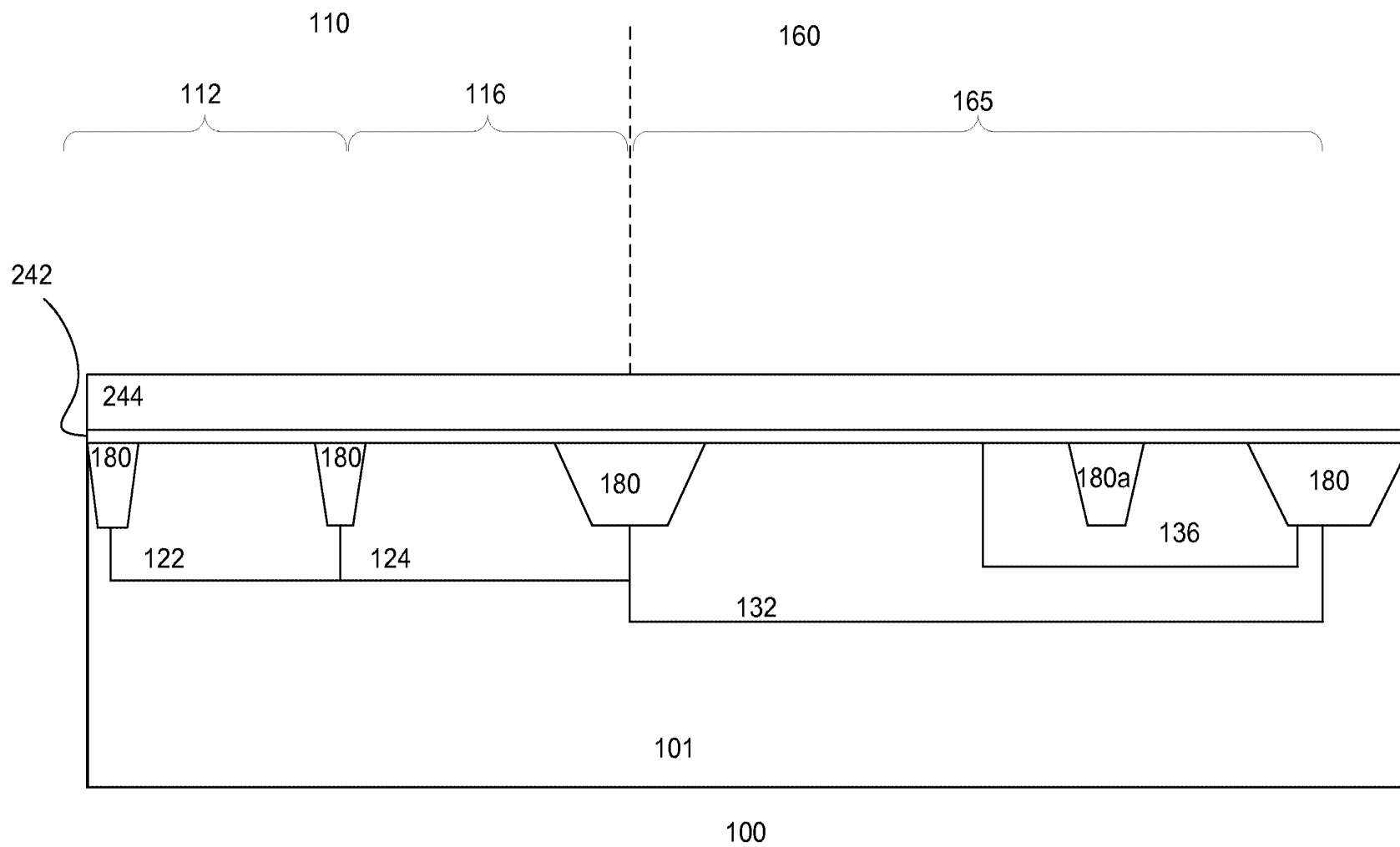


Fig. 2c

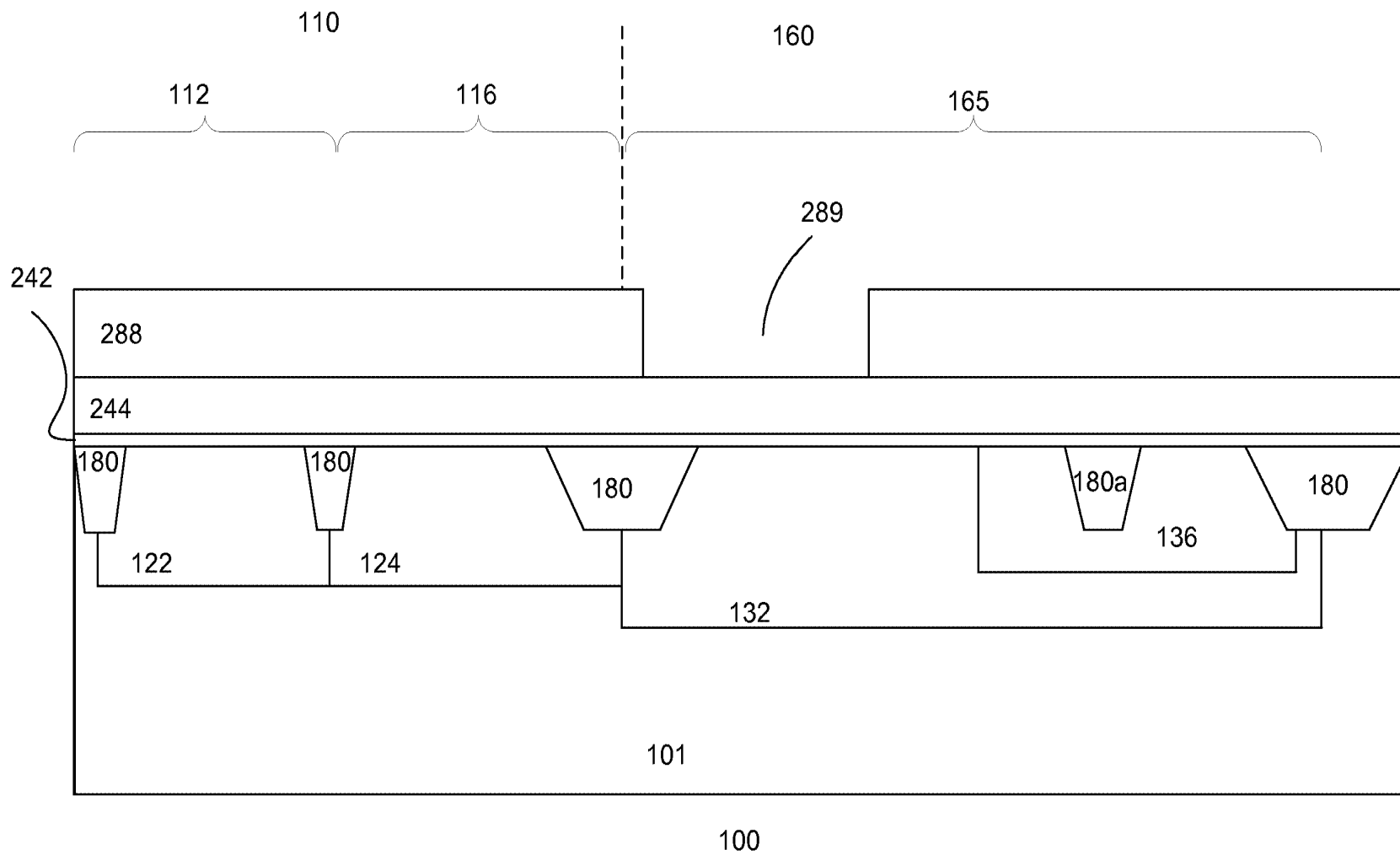


Fig. 2d

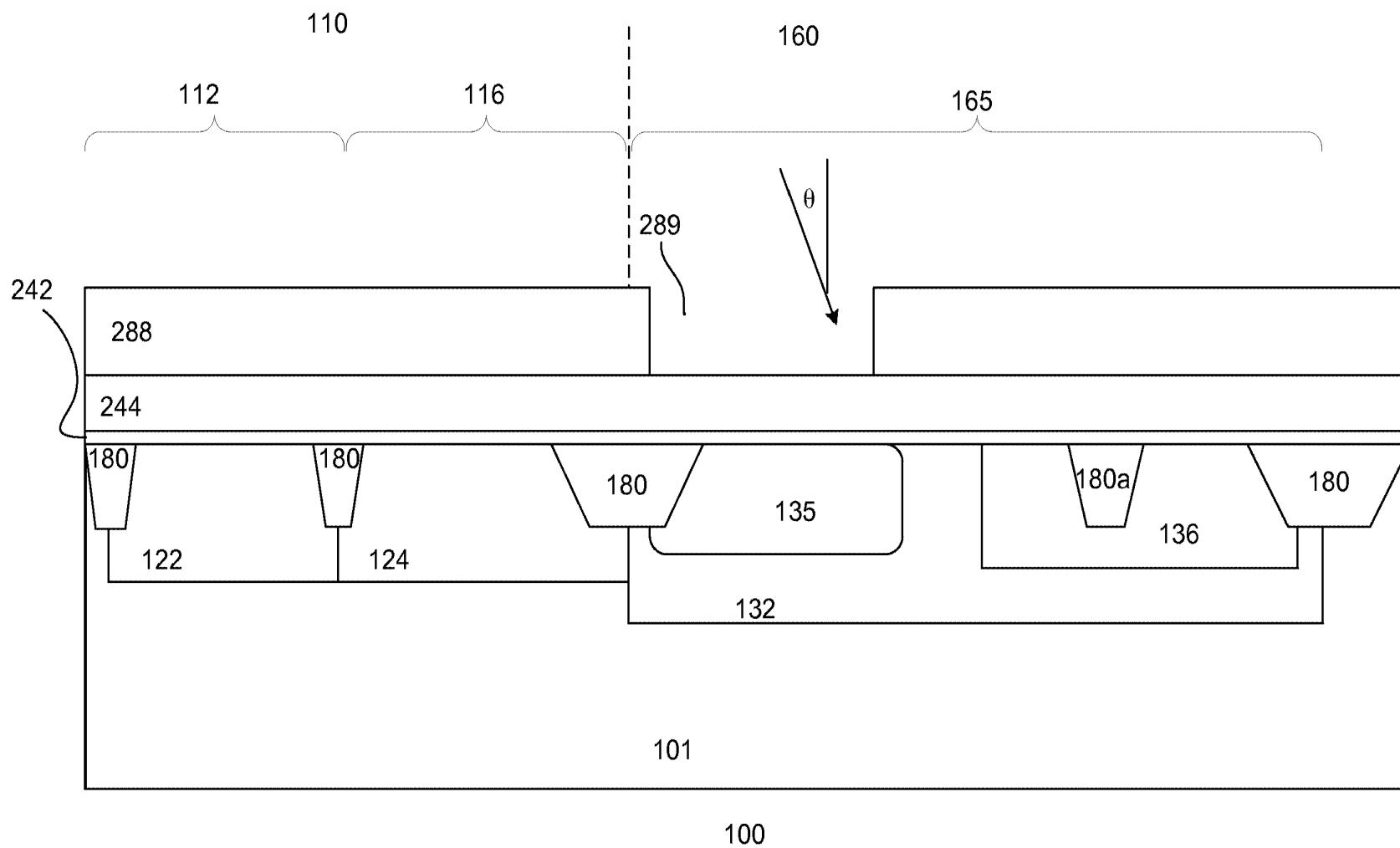


Fig. 2e

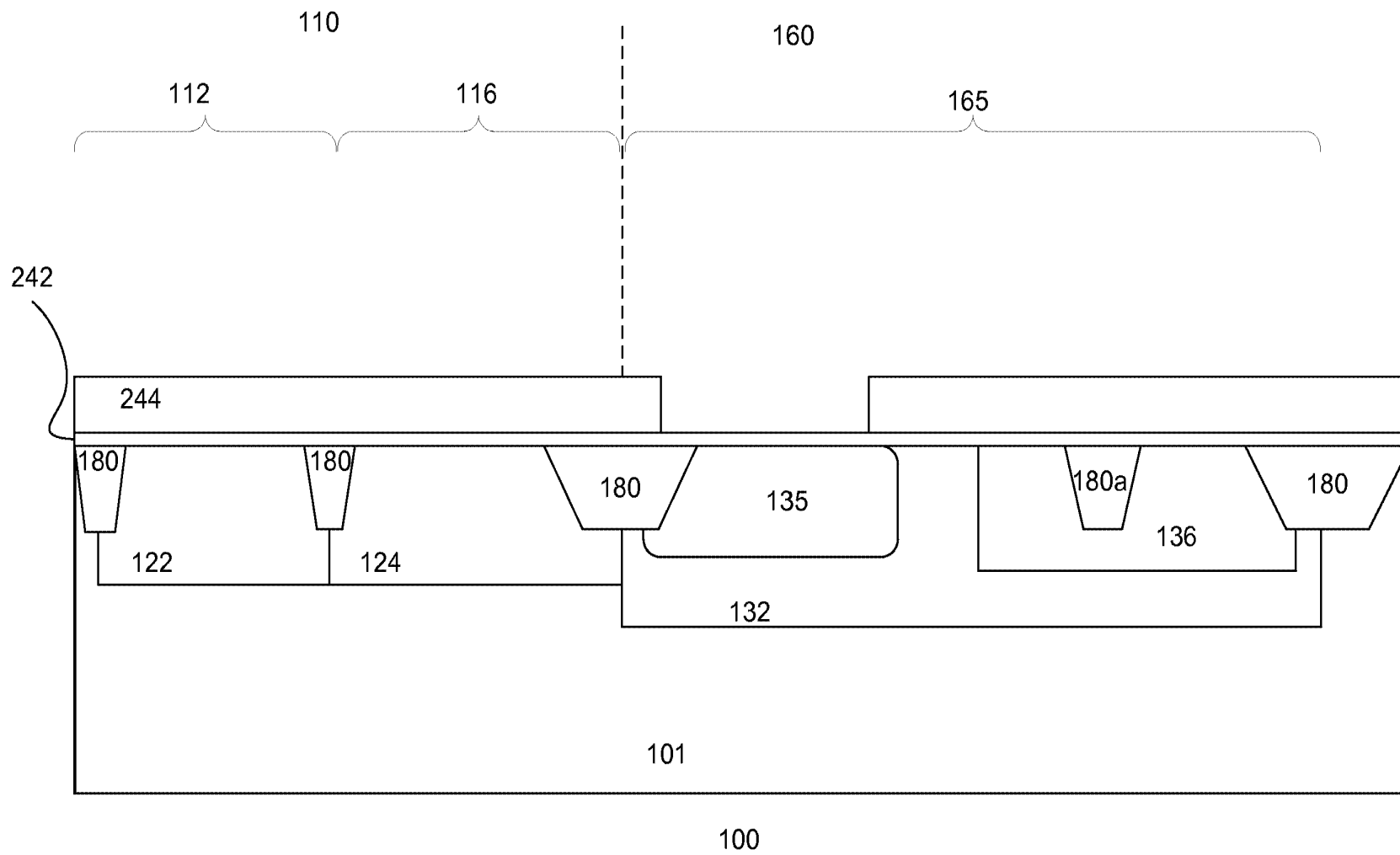


Fig. 2f

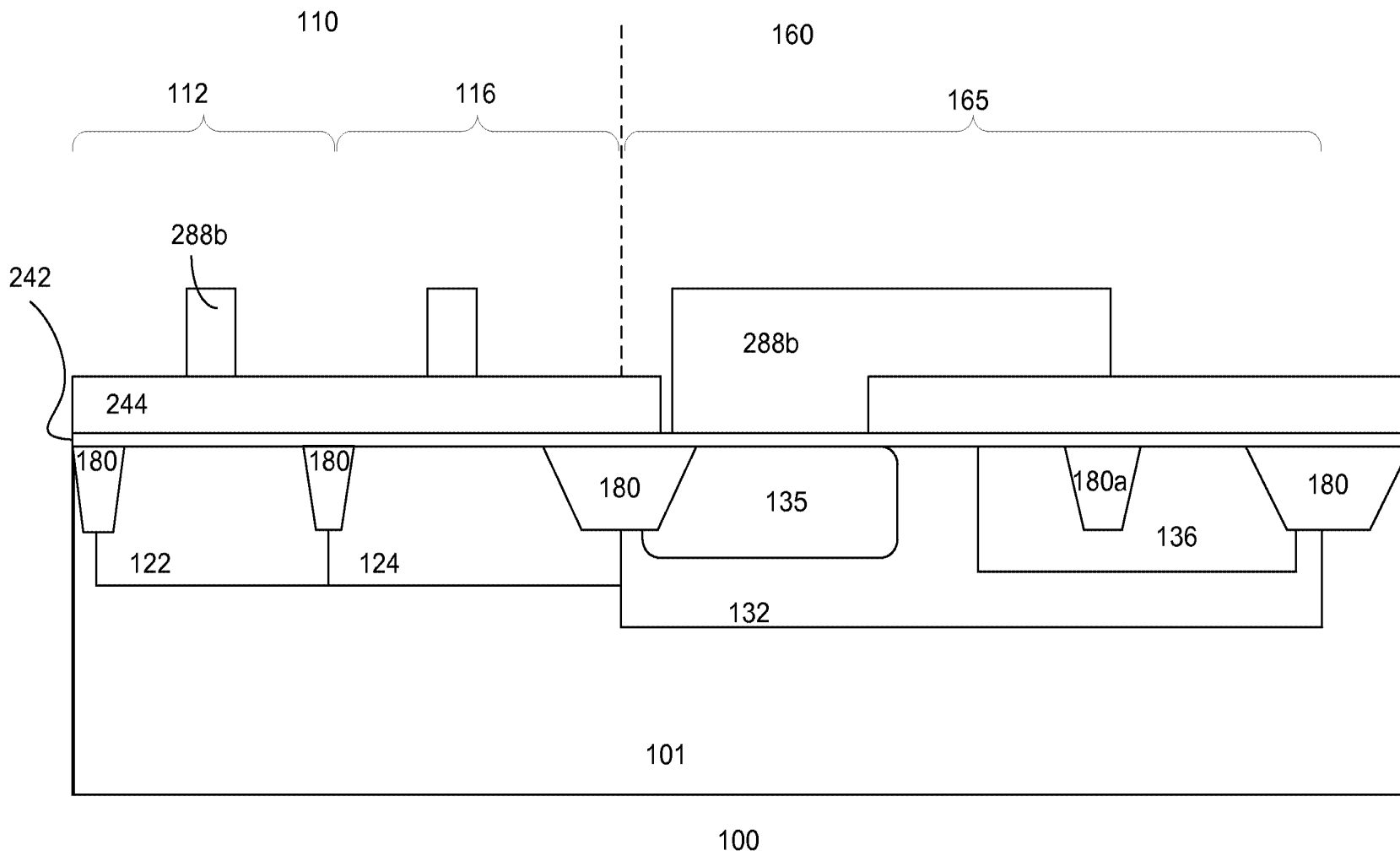


Fig. 2g

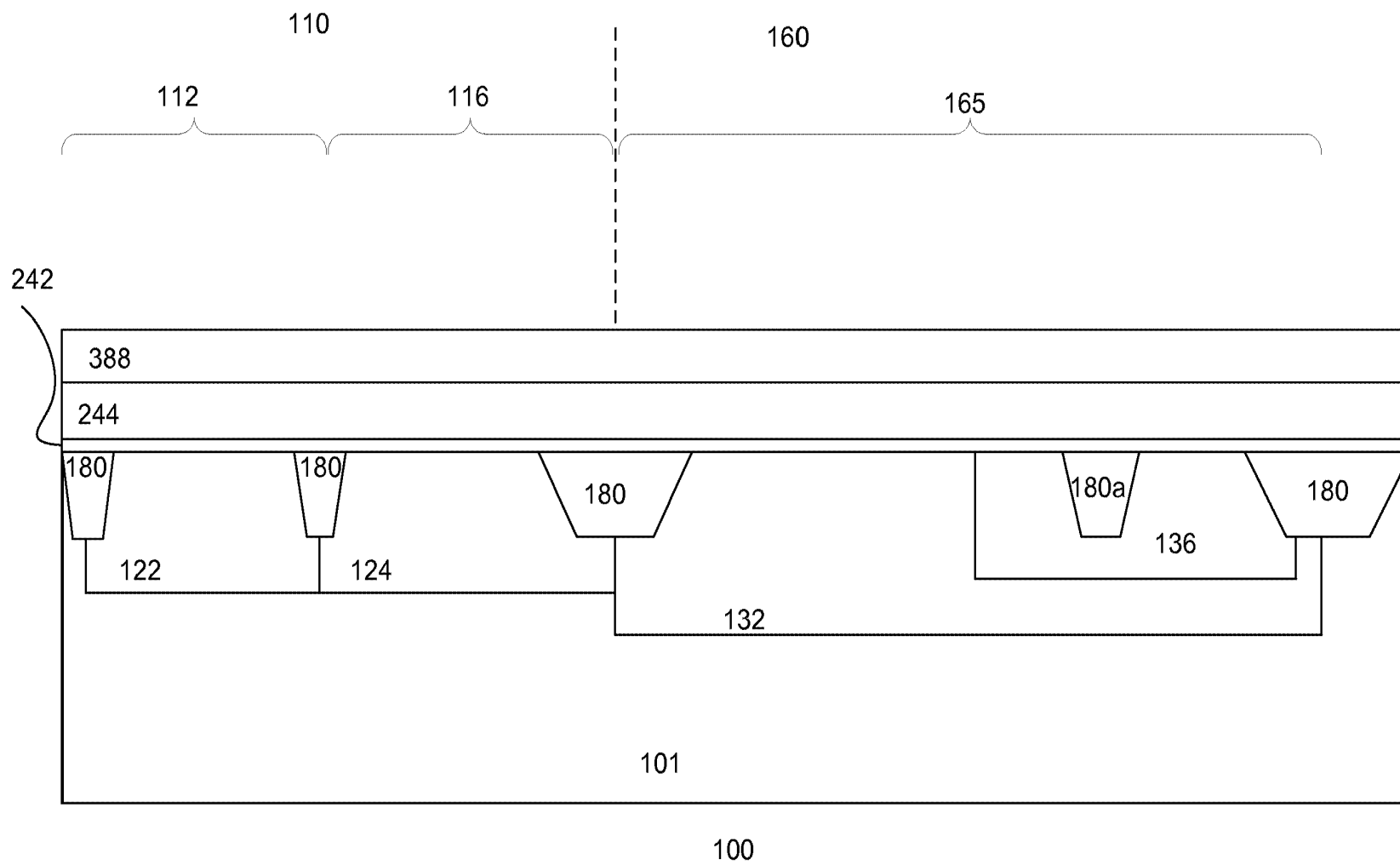


Fig. 3a

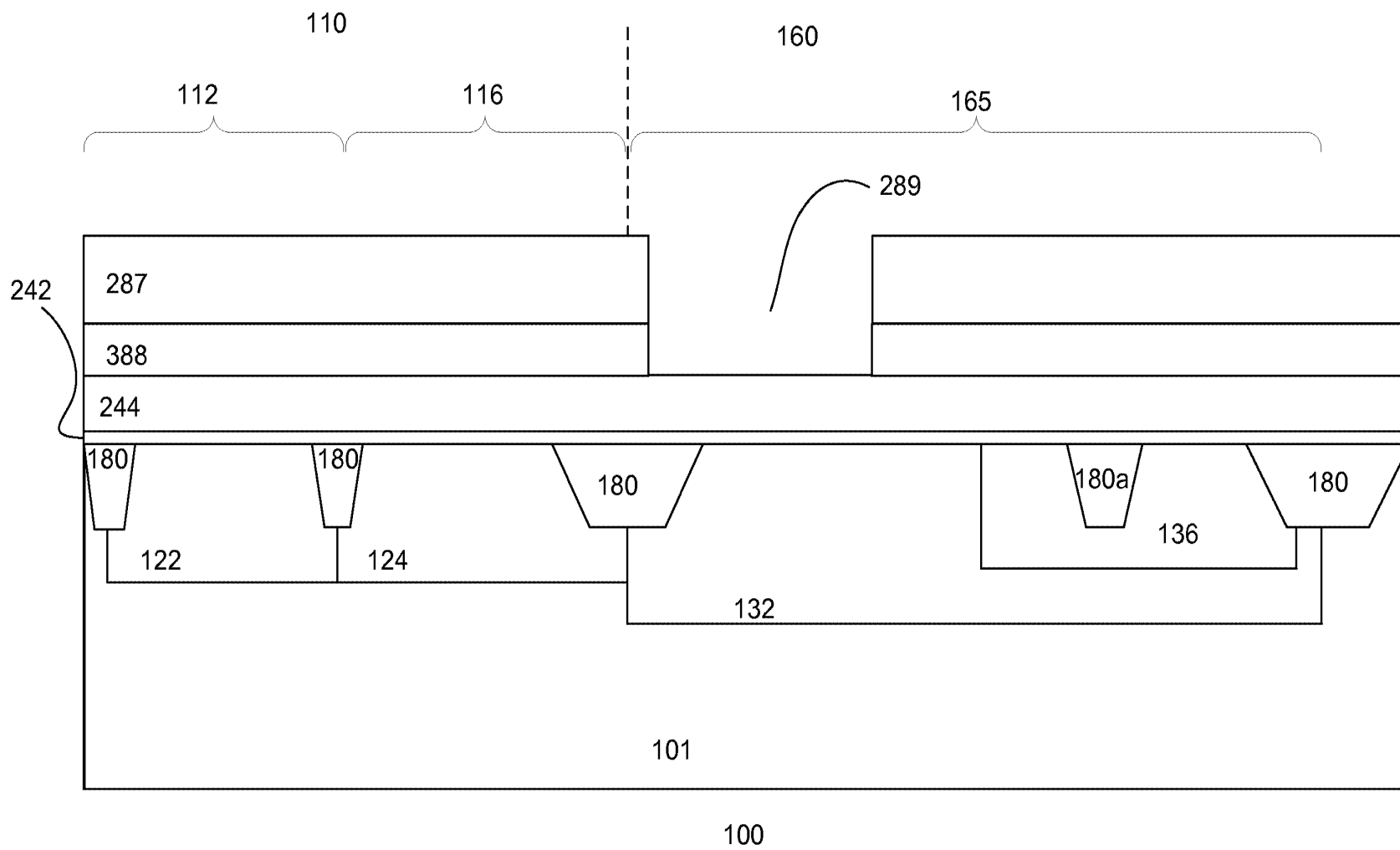


Fig. 3b

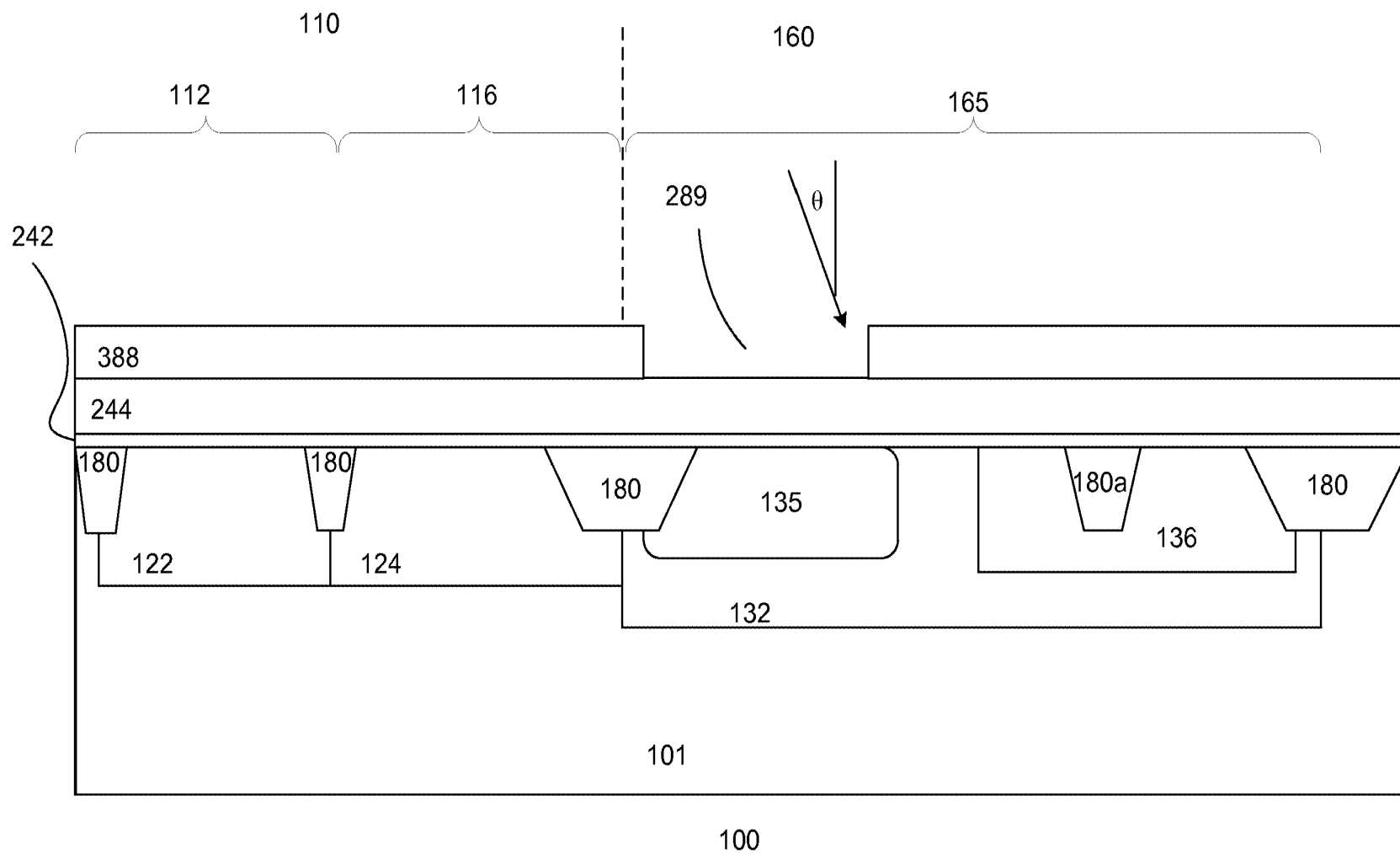


Fig. 3c

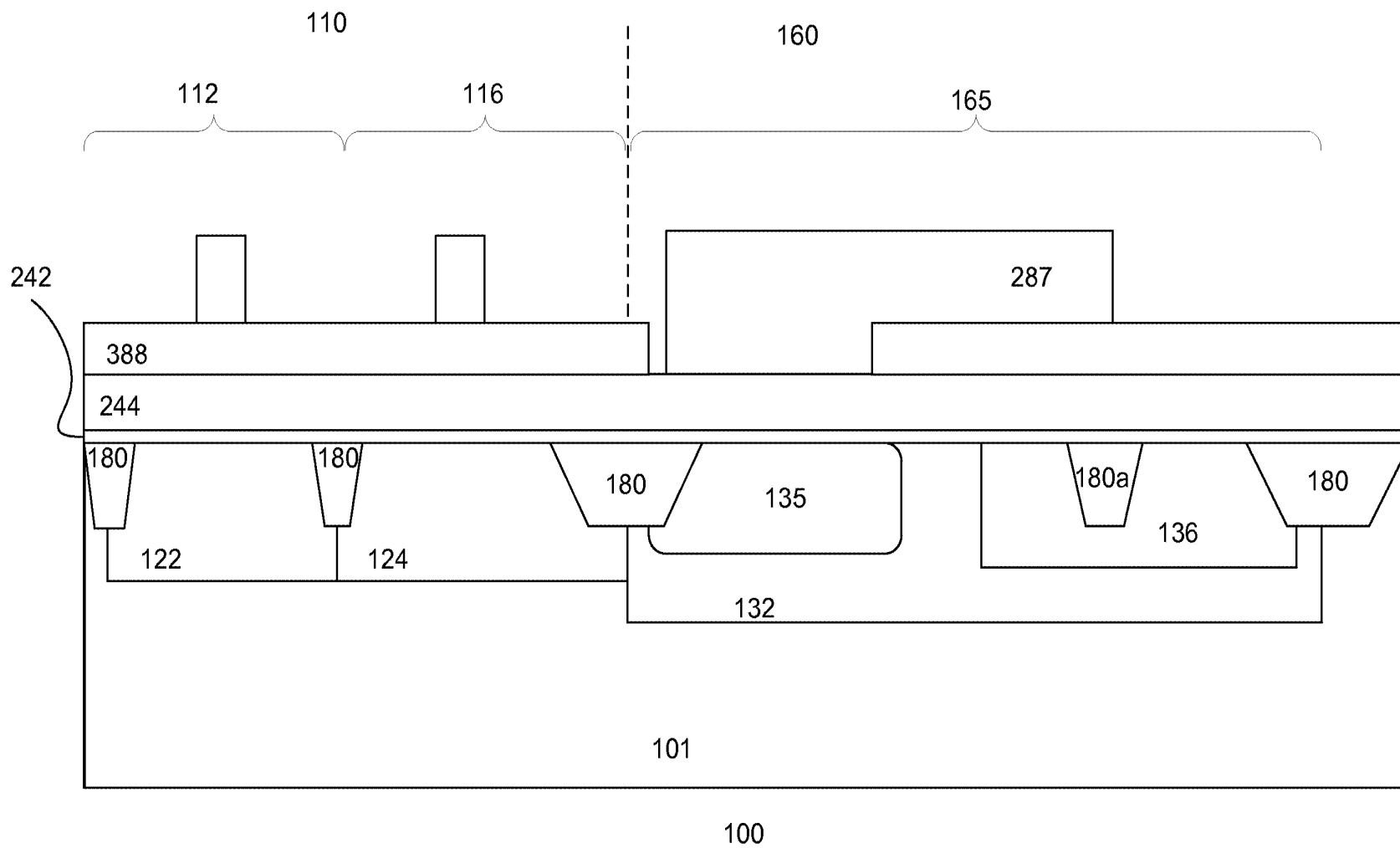


Fig. 3d

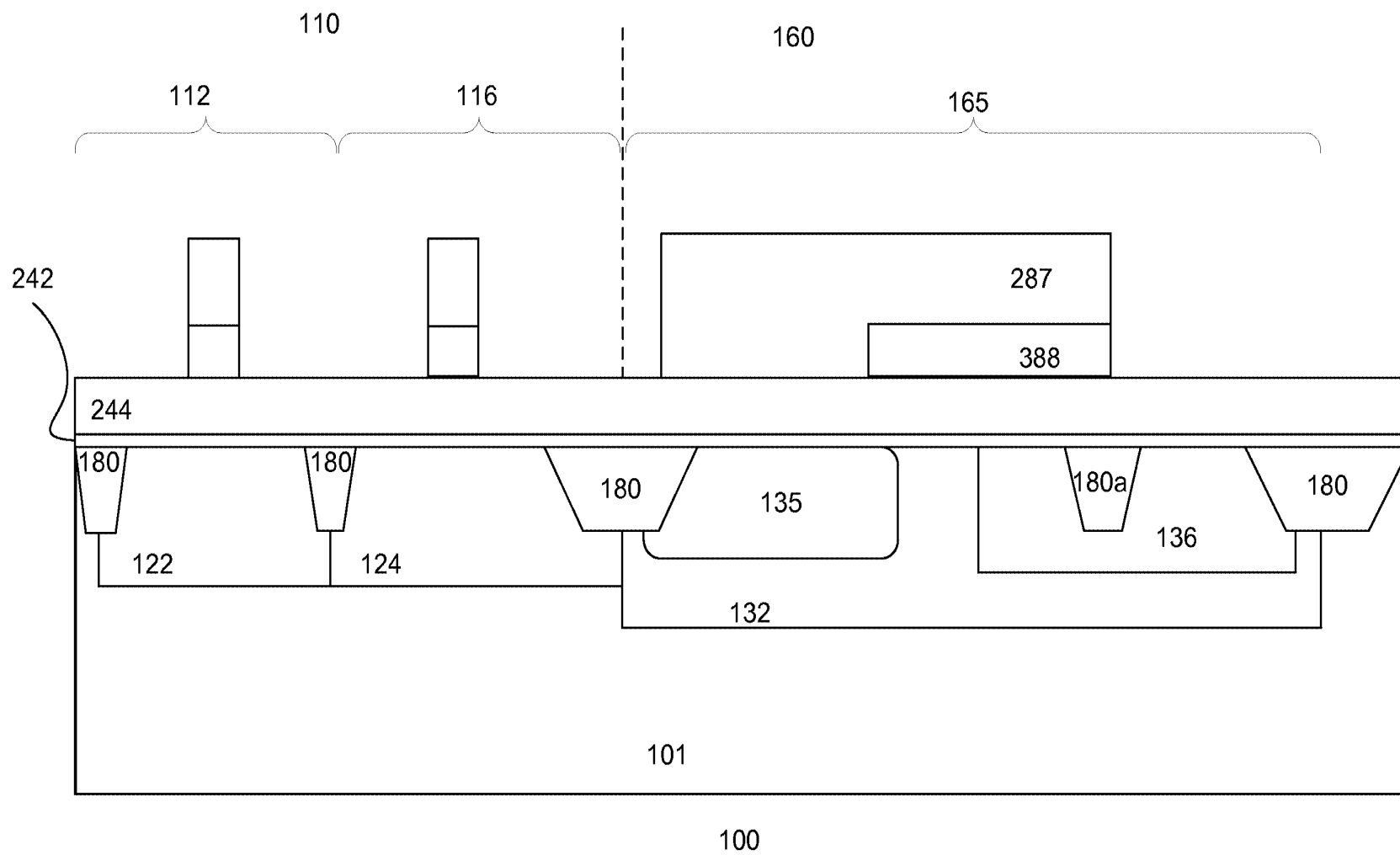


Fig. 3e

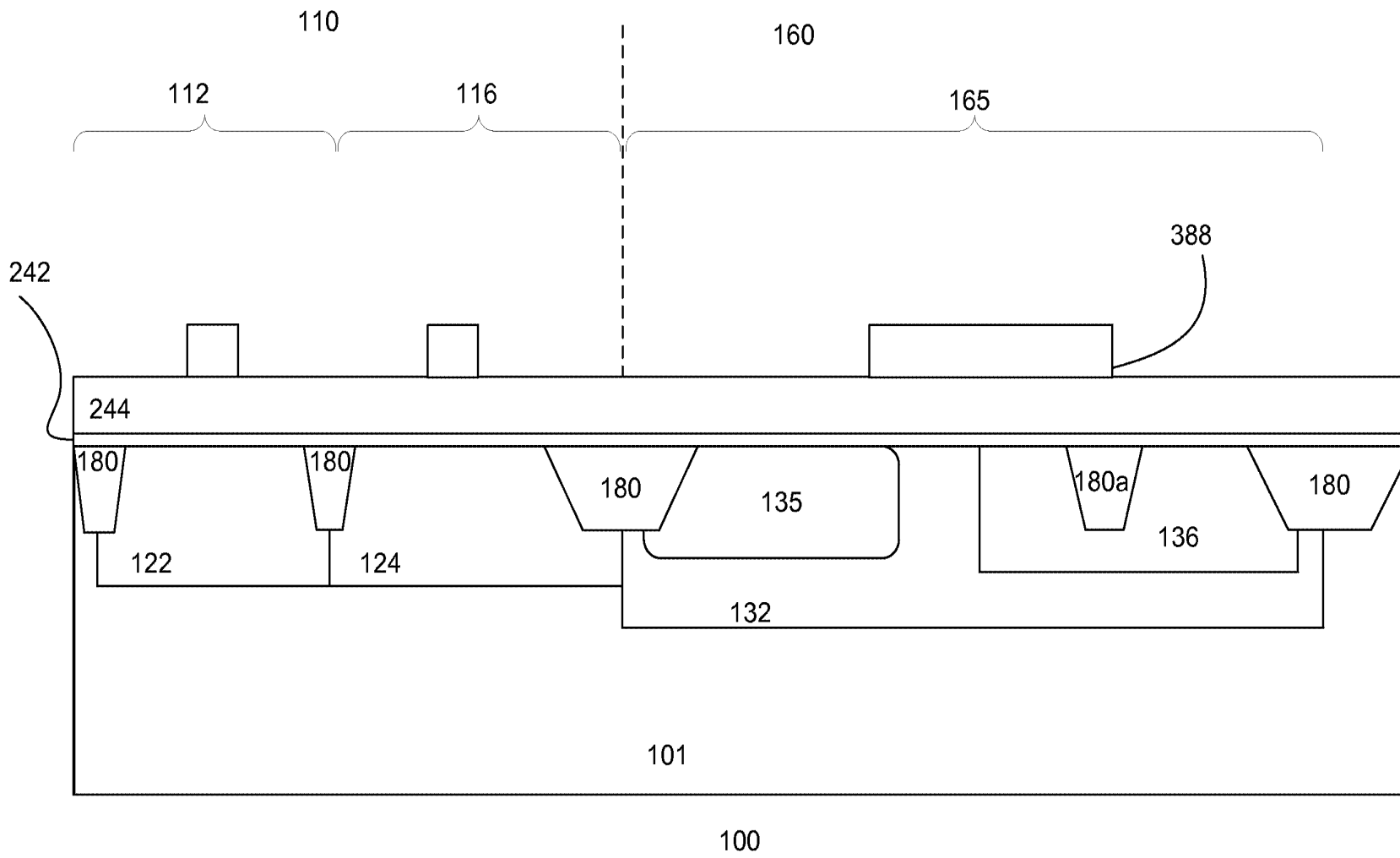


Fig. 3f

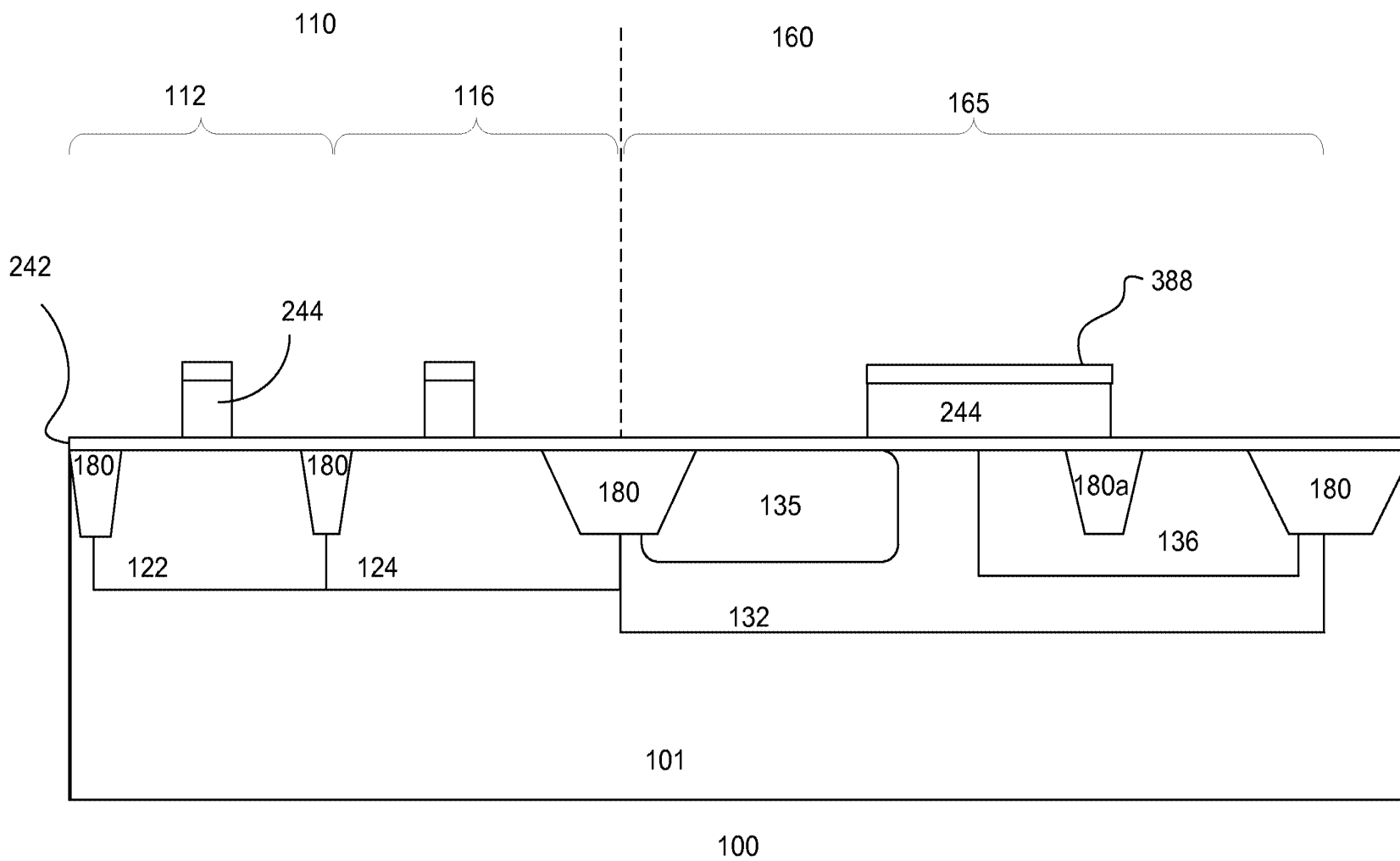


Fig. 3g

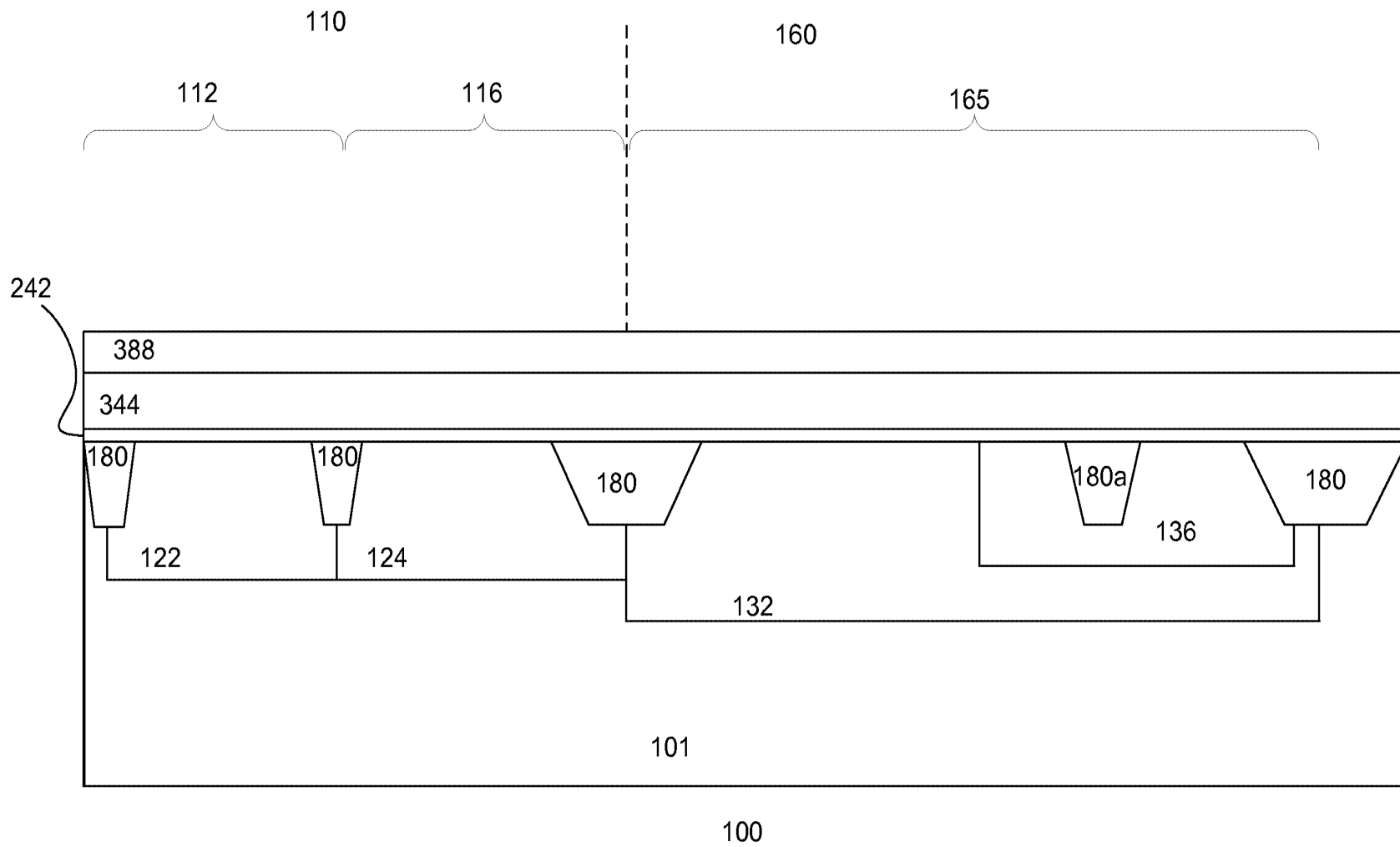


Fig. 4a

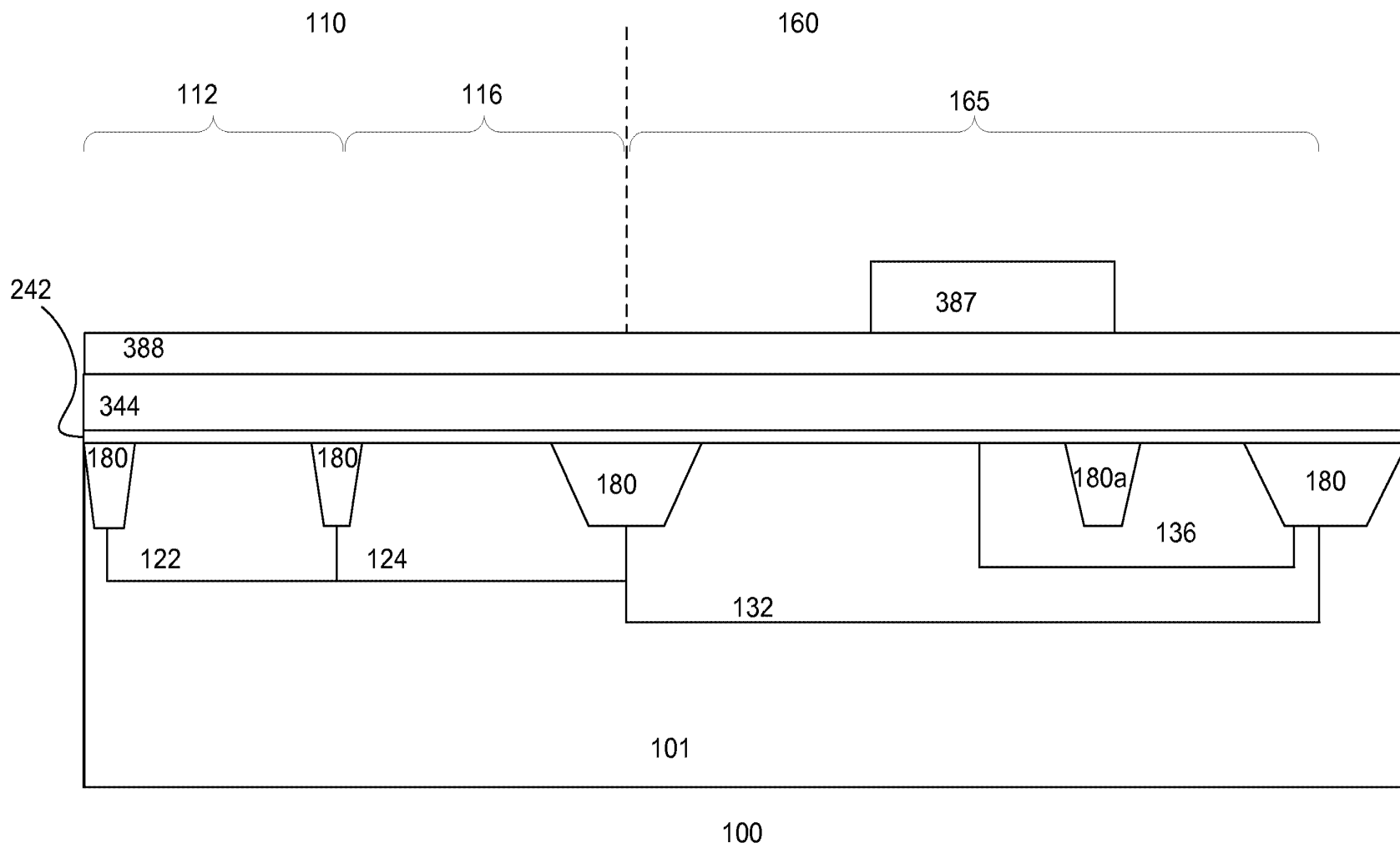


Fig. 4b

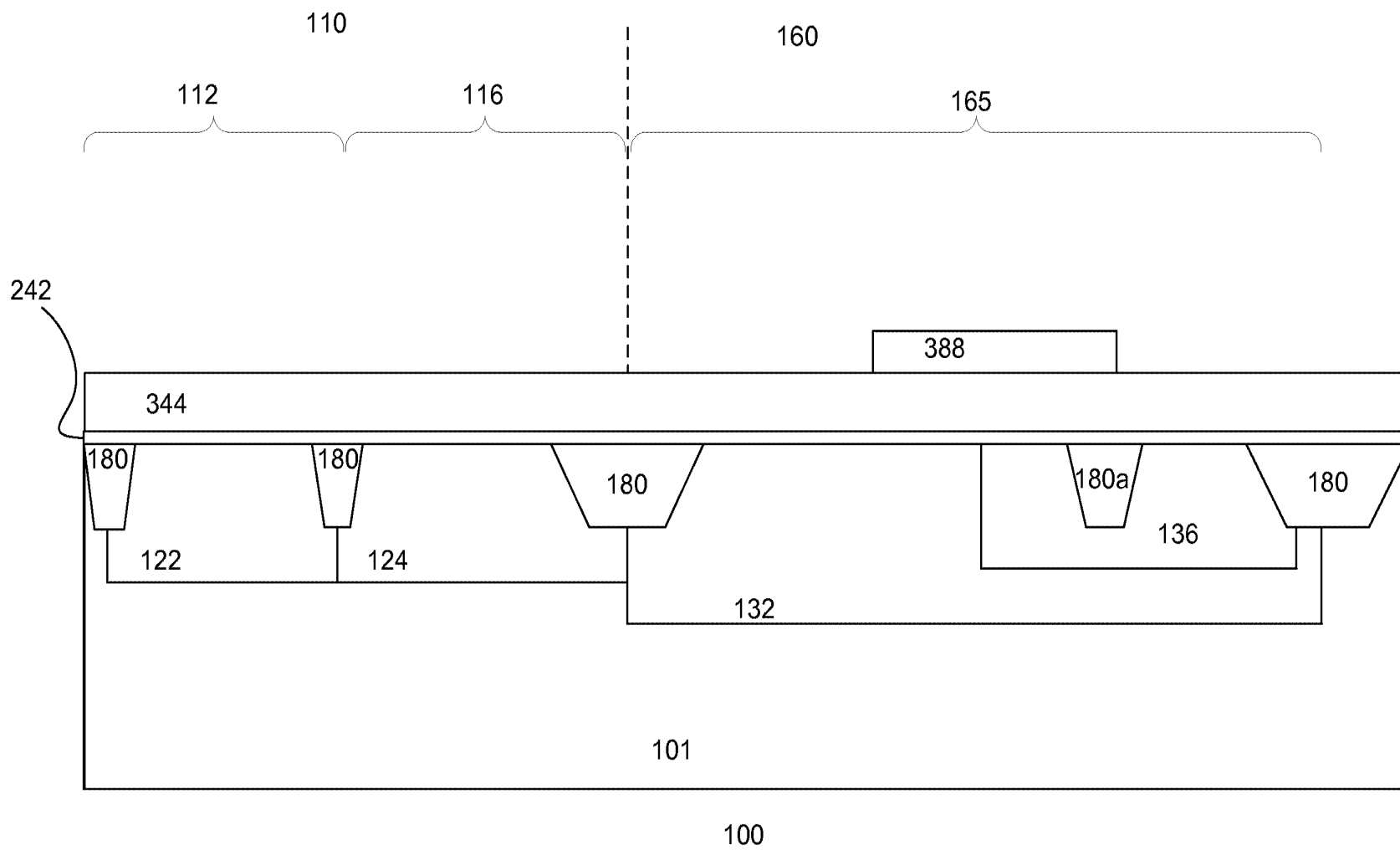


Fig. 4c

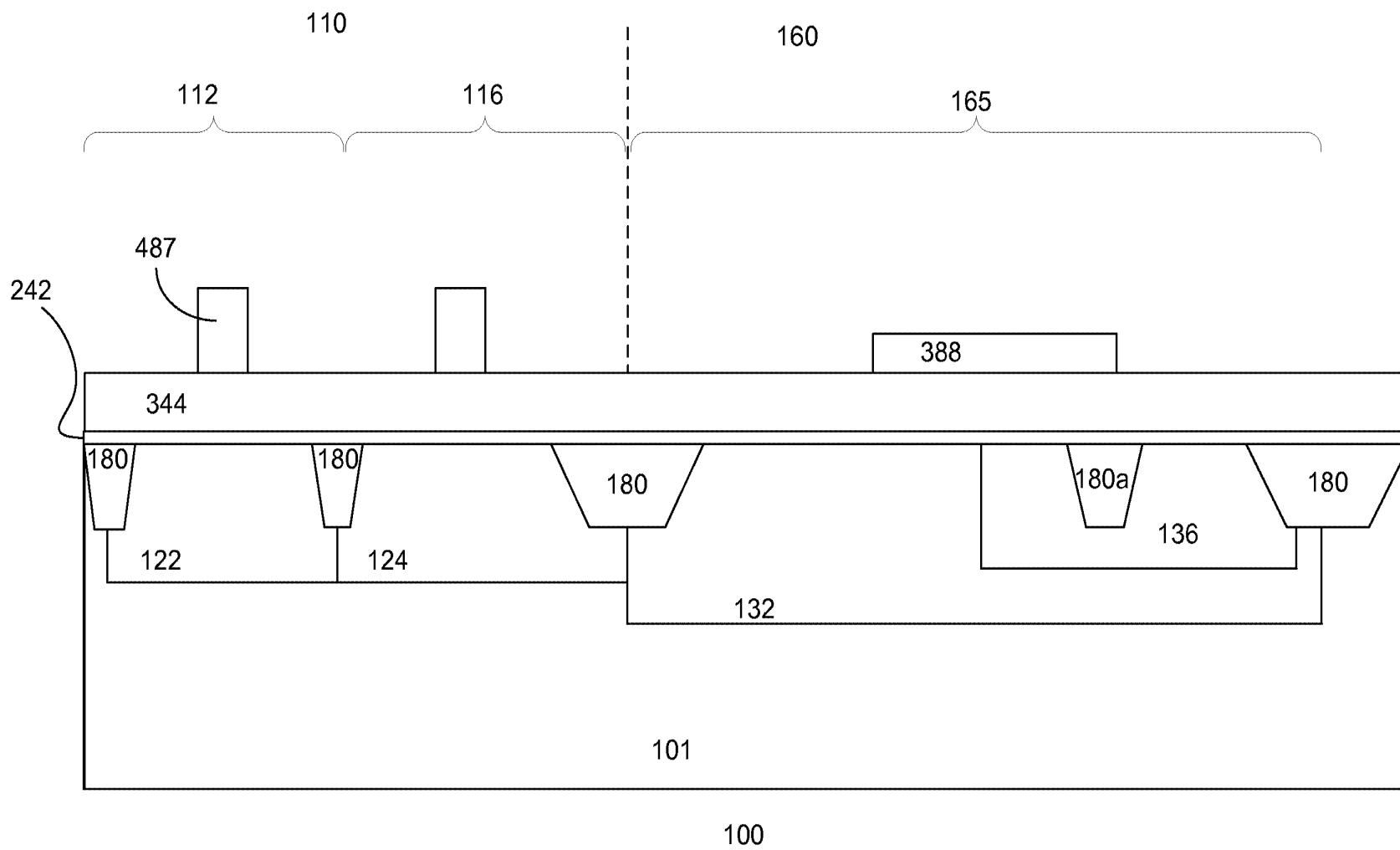


Fig. 4d

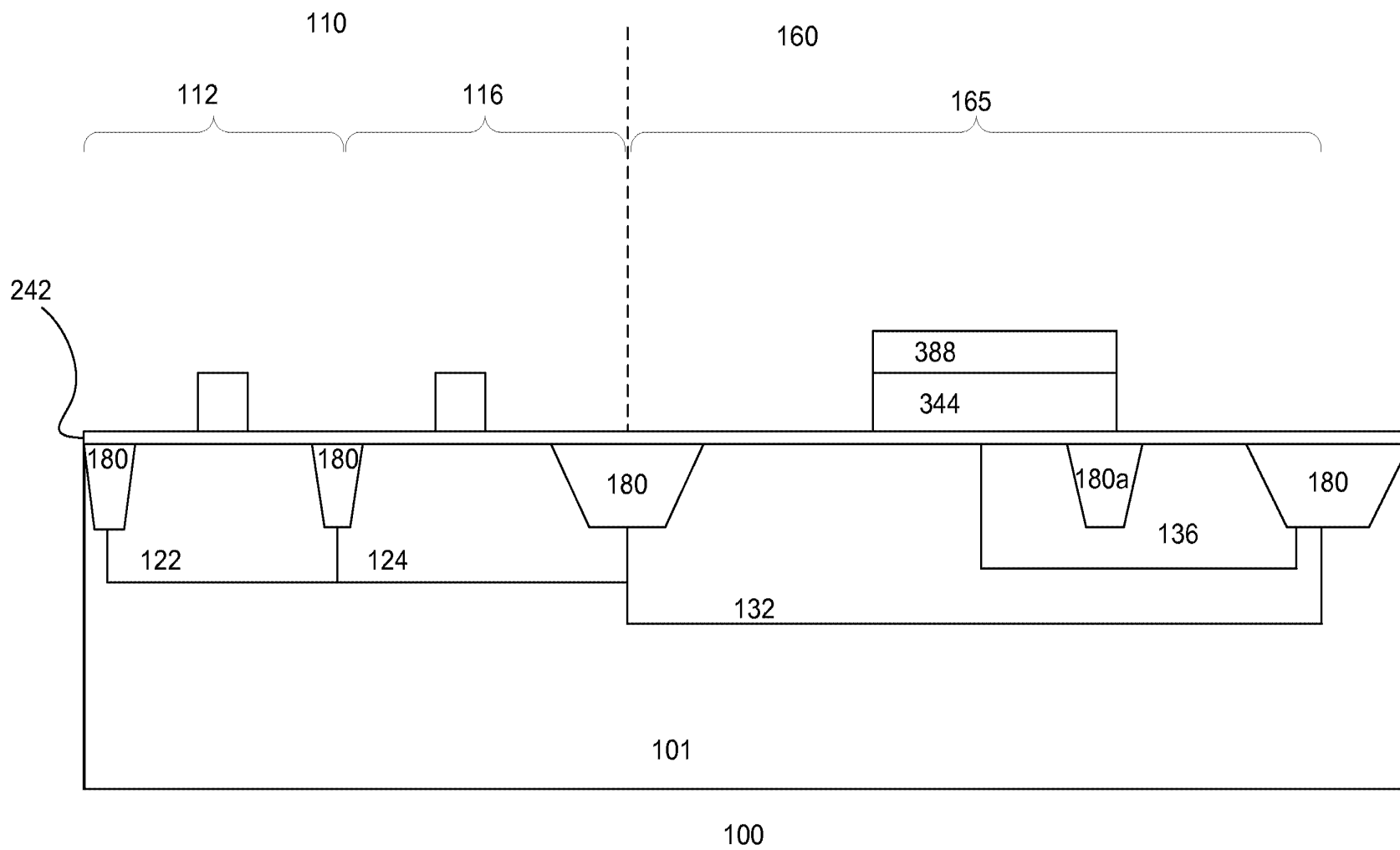


Fig. 4e

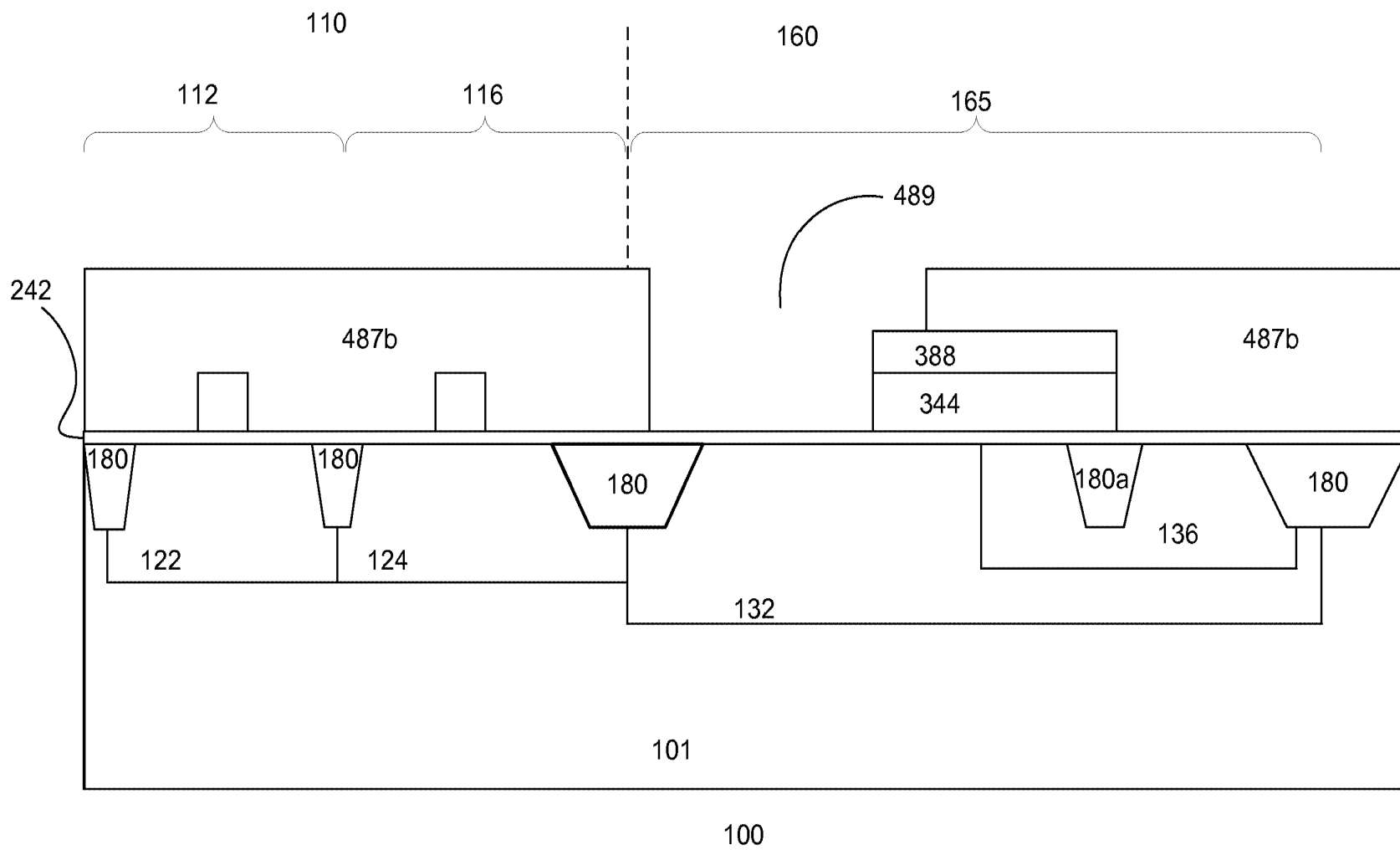


Fig. 4f

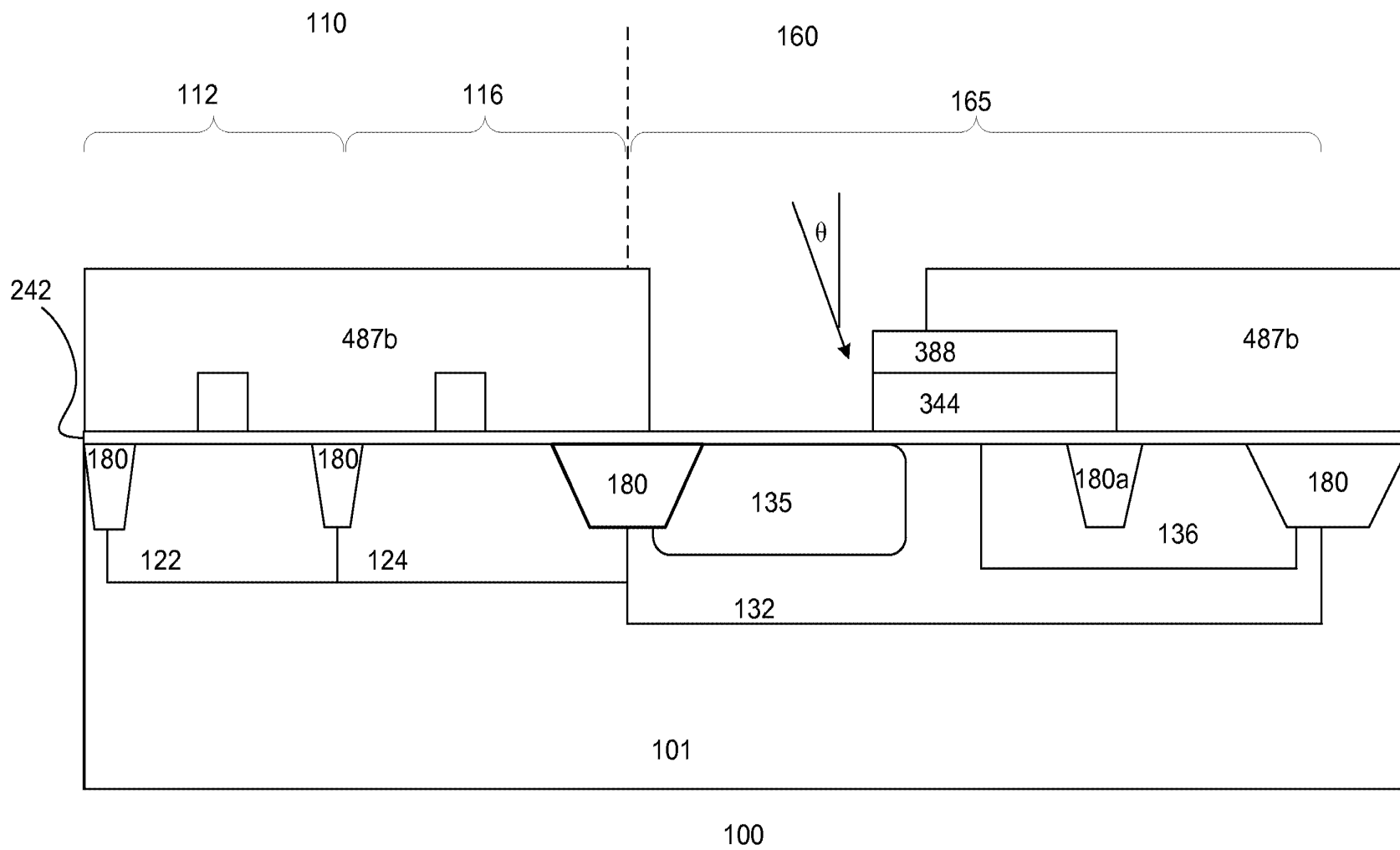


Fig. 4g

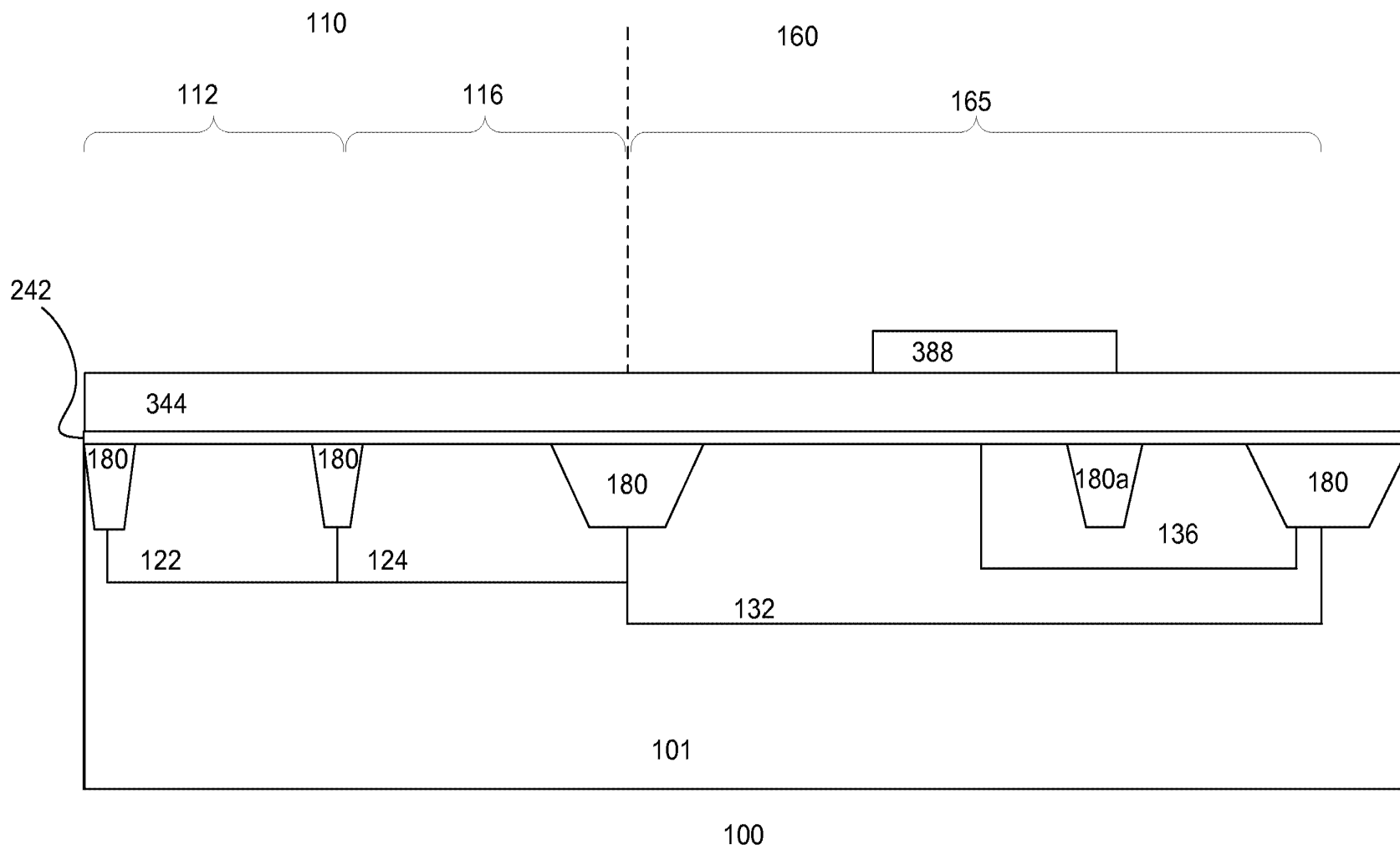


Fig. 5a

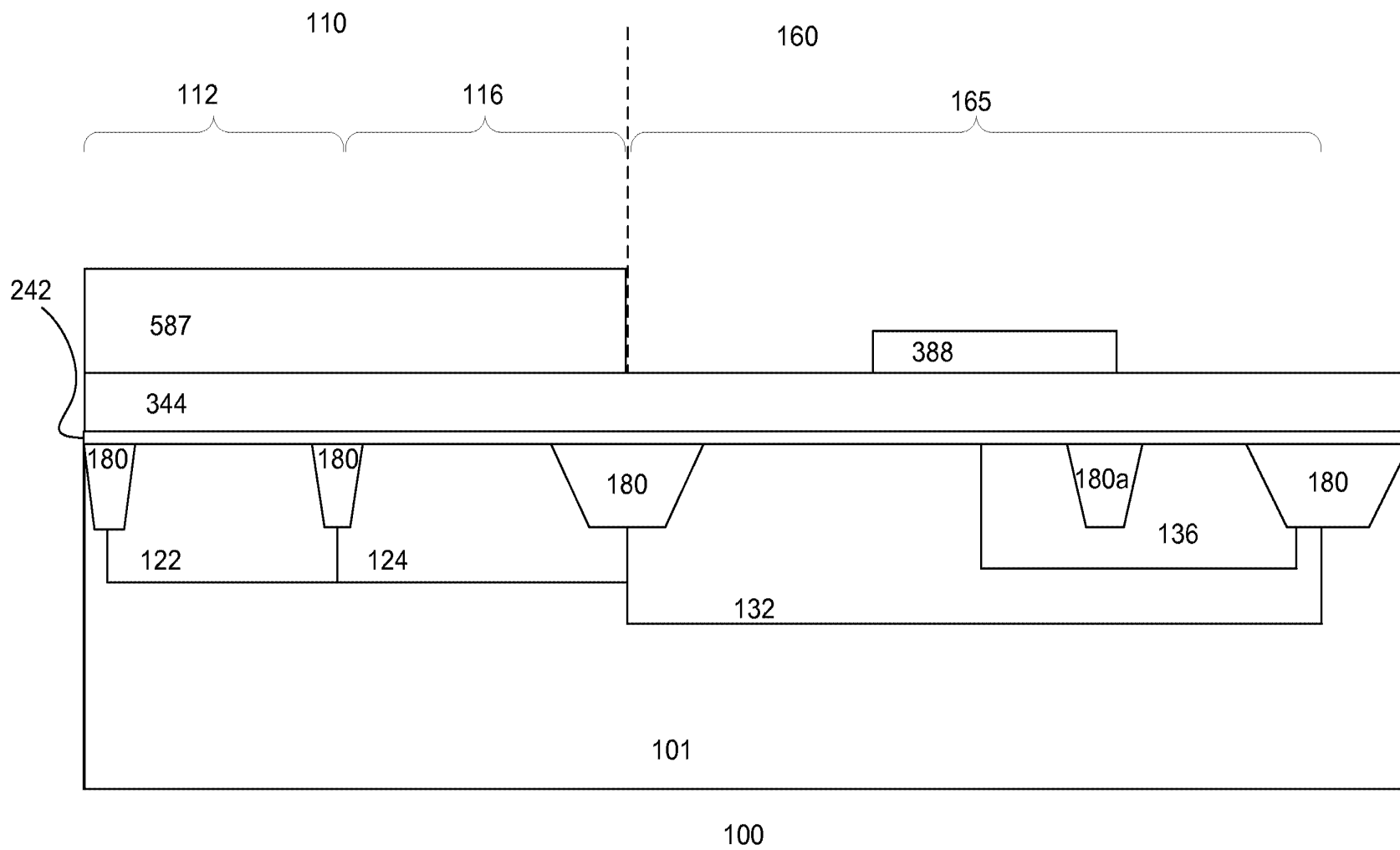


Fig. 5b

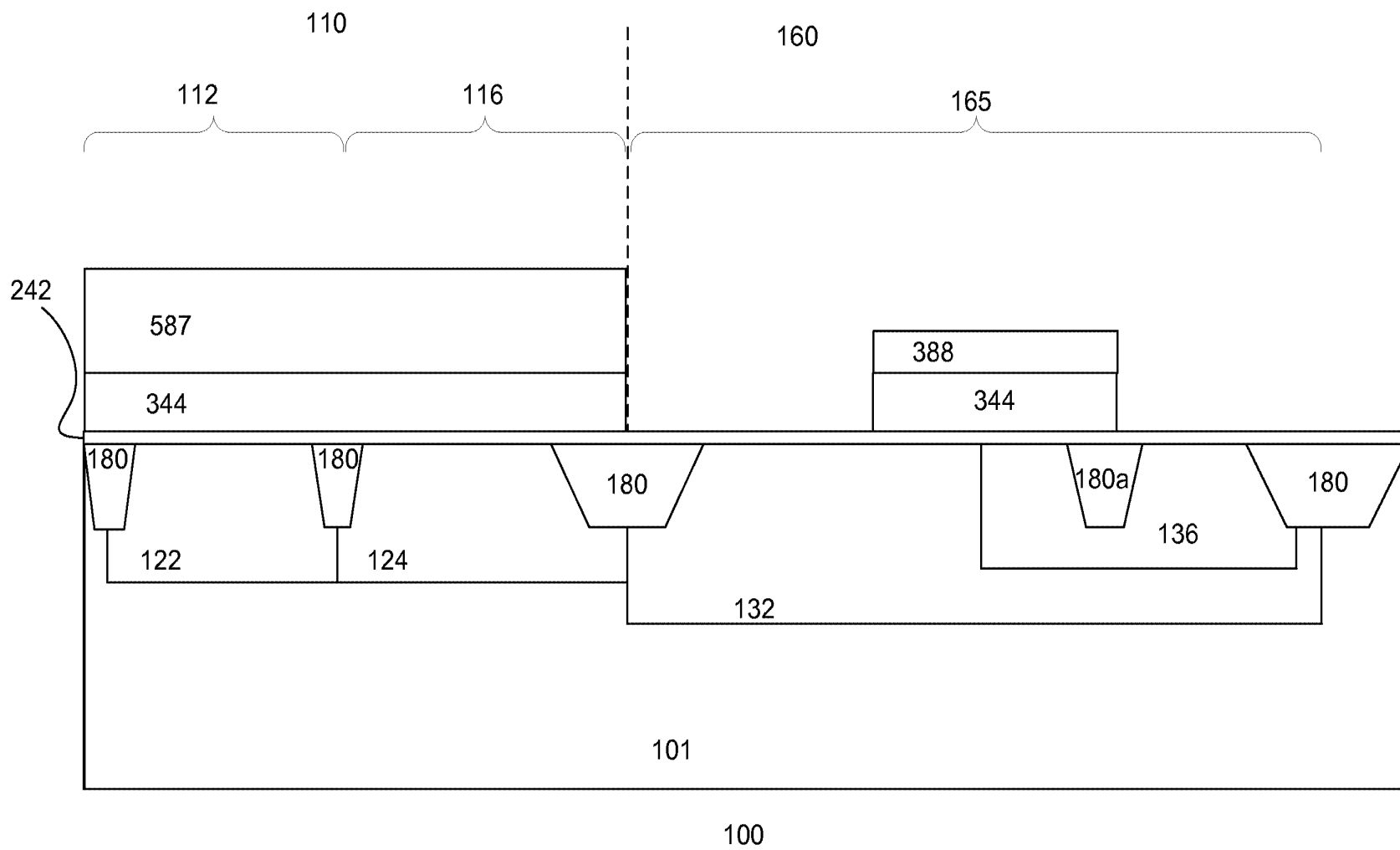


Fig. 5c

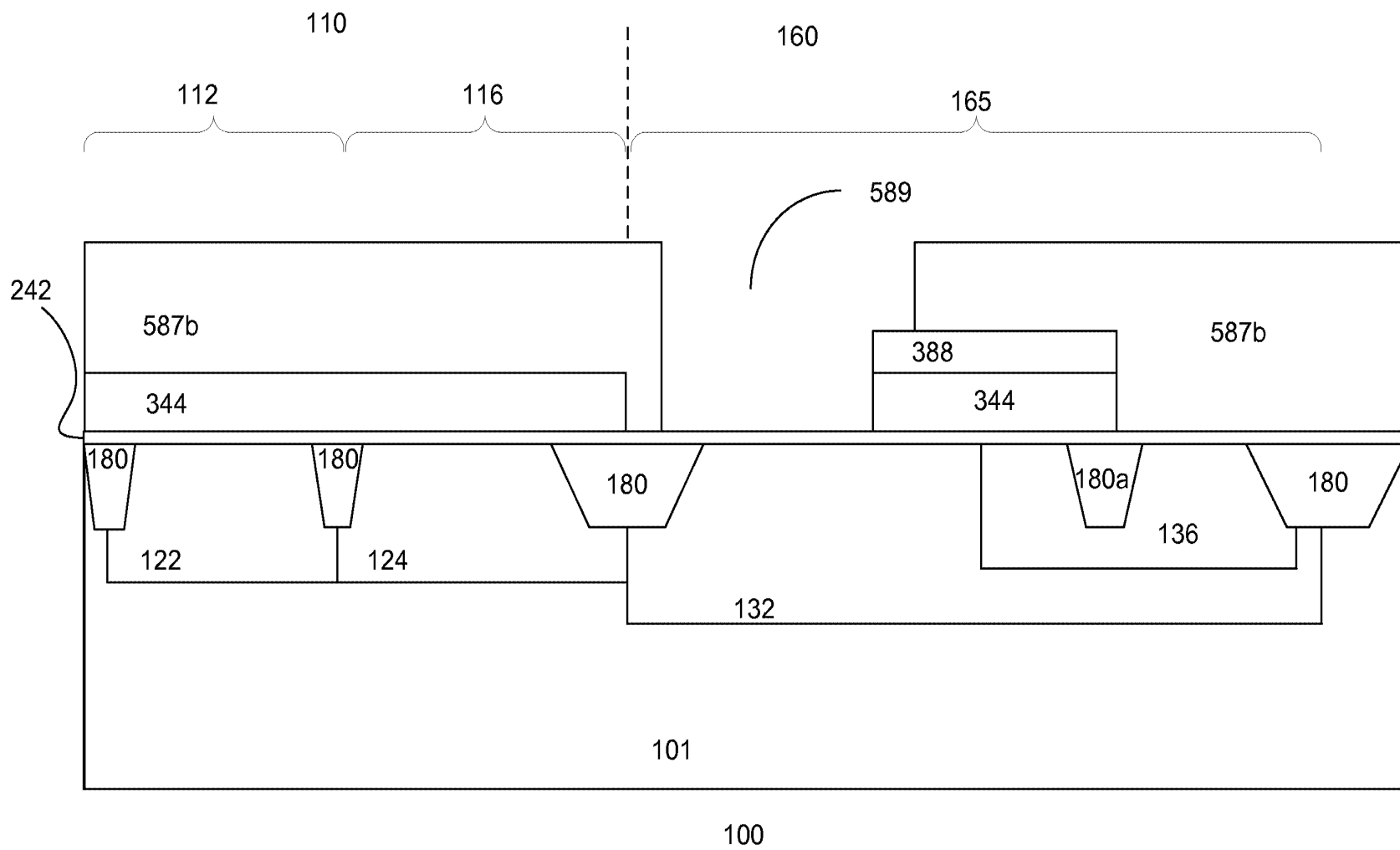


Fig. 5d

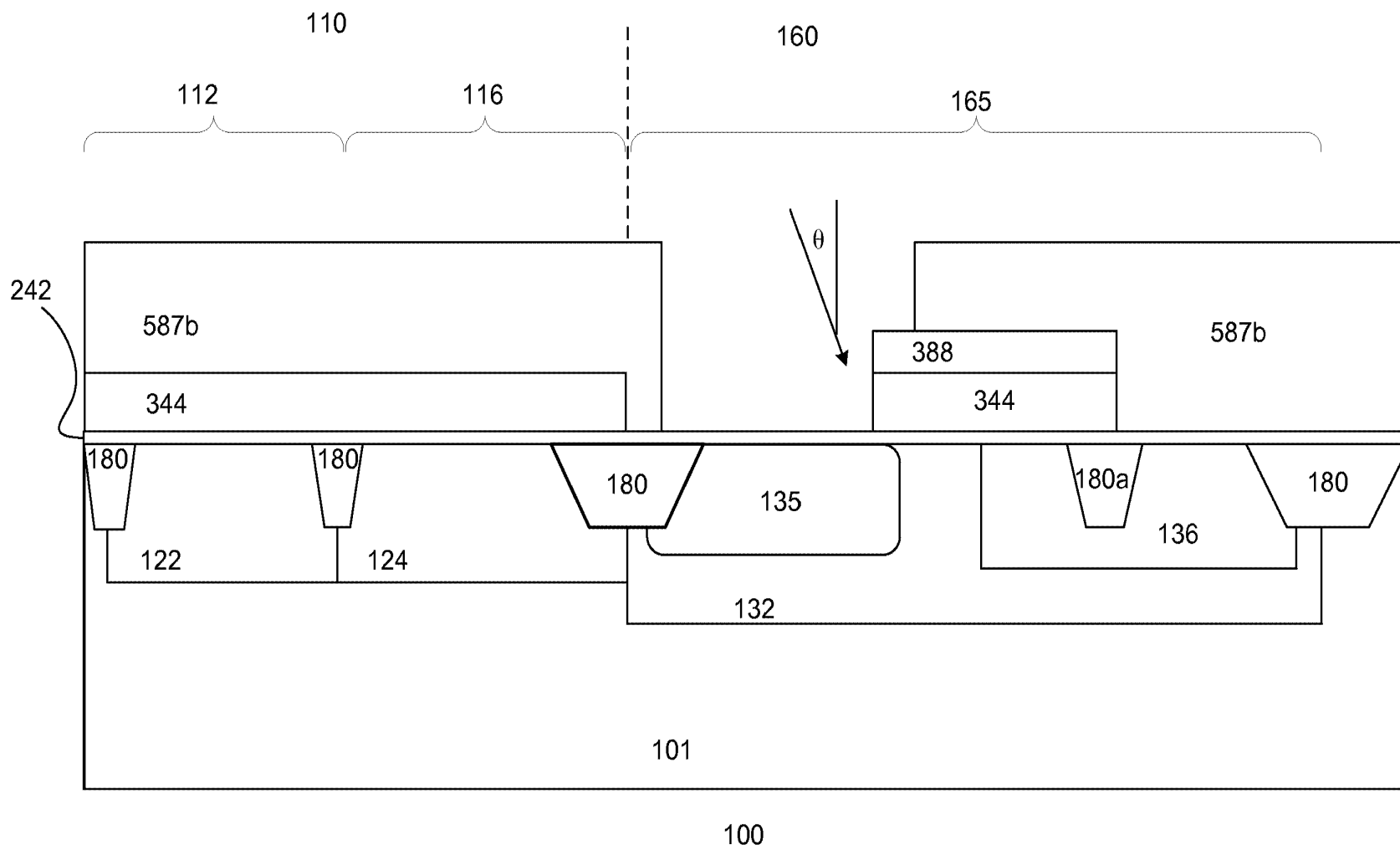


Fig. 5e

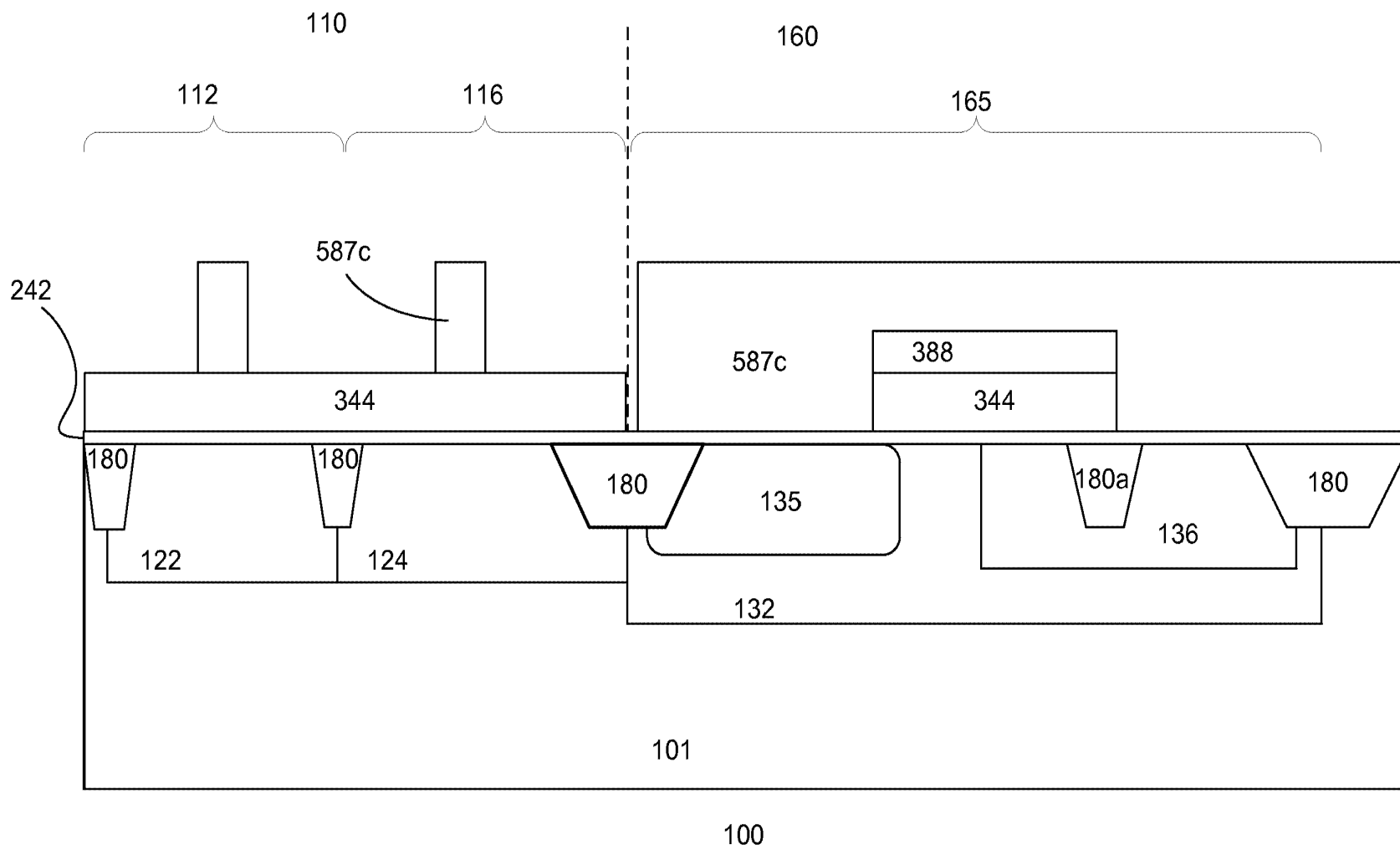


Fig. 5f

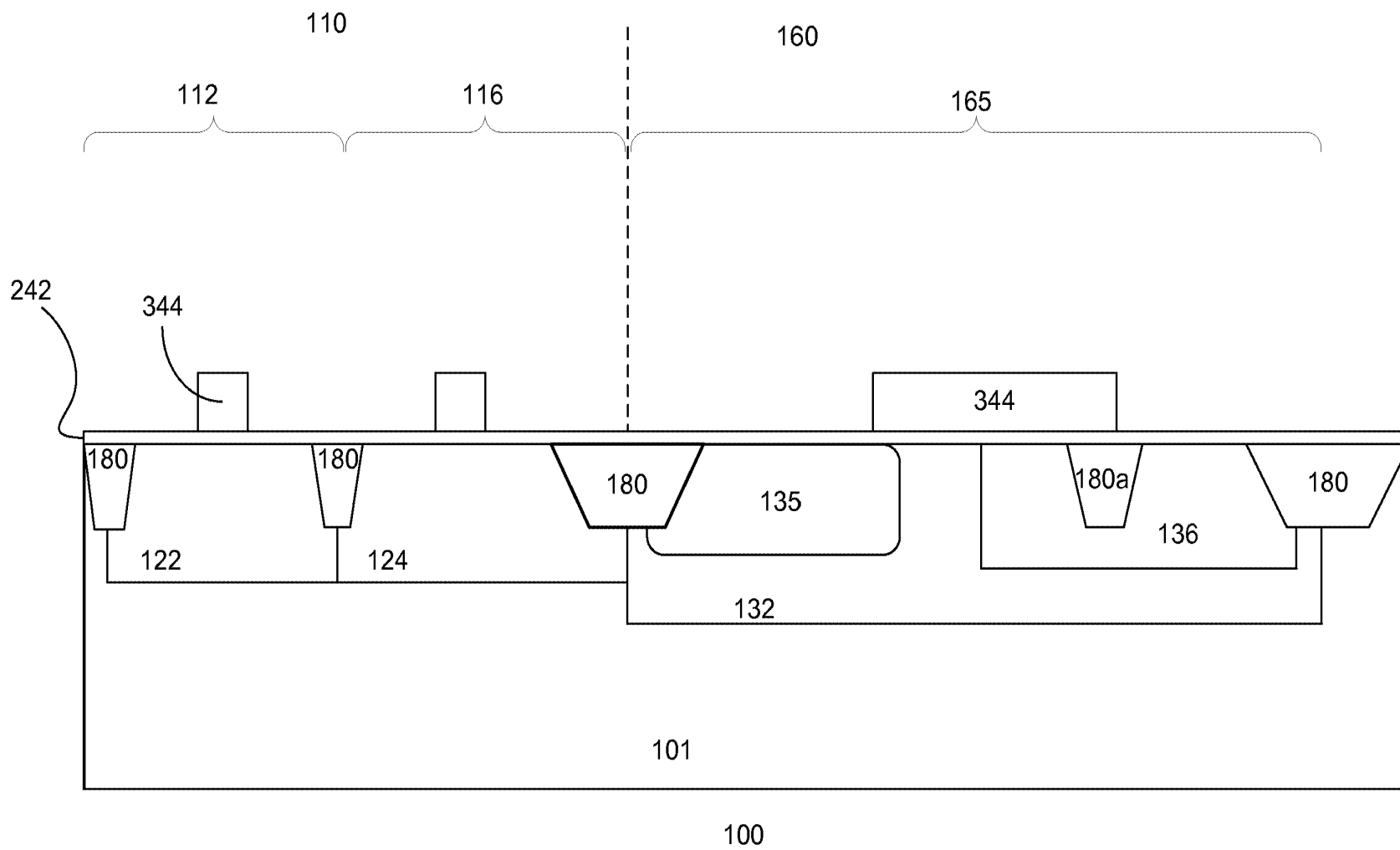


Fig. 5g

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Guowei ZHANG	
	Art Unit		
	Examiner Name	Not Assigned Yet	
	Attorney Docket Number	CSM P 2009 NAT 03 US 2	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	7776700		2010-08-17	Yang et al.	
	2	5589414		1996-12-31	Wann et al.	
	3	6759675		2004-07-06	Csutak et al.	
	4	7180132		2007-02-20	Cai et al.	
	5	6894350		2005-05-17	Shimizu et al.	
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Guowei ZHANG	
Art Unit		
Examiner Name	Not Assigned Yet	
Attorney Docket Number	CSM P 2009 NAT 03 US 2	

	1	20090059111		2009-03-05	JANG	
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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² ;	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

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NON-PATENT LITERATURE DOCUMENTS

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	LEONARD RUBIN ET AL., Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, 01 July 2003, pages 119-125, Volume 46 Issue 7, Solid State Technology, USA.	☐

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Guowei ZHANG
Art Unit	
Examiner Name	Not Assigned Yet
Attorney Docket Number	CSM P 2009 NAT 03 US 2

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/dexter chin/	Date (YYYY-MM-DD)	2012-07-16
Name/Print	Dexter CHIN	Registration Number	38,842

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

DECLARATION AND POWER OF ATTORNEY

As a below named inventor, I hereby declare that:

My residence, post office address, and citizenship are as stated below next to my name,

I believe I am the original, first and sole inventor (if only one name is listed below) or an original, first and joint inventor (if plural names are listed below) of the subject matter claimed and for which a patent is sought on the invention entitled:

HIGH VOLTAGE DEVICE

the specification of which

☒ is attached hereto

☐ was filed on ___ as Application Serial No. ___ and was amended on ___ (if applicable).

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claims, as amended by any amendment referred to above.

I do not know and do not believe the same was ever known or used in the United States of America before this invention thereof or more than one year prior to this application, and that the same was not in public use or on sale in the United States of America more than one year prior to this application. I acknowledge the duty to disclose information which is known to me to be material to patentability in accordance with title 37, Code of Federal Regulations, section 1.56.

I hereby claim foreign priority benefits under Title 35, United States Code, Section 119(a)-(d) or Section 365(b) of any foreign application(s) for patent or inventor's certificate, or Section 365(a) of any PCT international application which designated at least one country other than the United States, listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed:

Prior Foreign Application(s):

<u>Number</u>	<u>Country</u>	<u>Day/Month/Year filed</u>	<u>Priority Claimed</u>	
			<u>Yes</u>	<u>No</u>
				☒

I hereby claim the benefit under 35 USC 119(e) of any United States provisional application(s) listed below:

Prior Provisional Application(s):

<u>Application Number</u>	<u>Filing Date</u>

I hereby claim the benefit under Title 35, United States Code, Section 120 of any United States application(s), or Section 365(c) of any PCT international application designating the United States, listed below and, insofar as the subject matter of each of the claims of this application is not disclosed in the prior United States or PCT international application in the manner provided by the first paragraph of Title 35, United States Code, Section 112, I acknowledge the duty to disclose material information as defined in Title 37, Code of Federal

Regulations, Section 1.56 which occurred between the filing date of the prior application and the national or PCT international filing date of this application:

Prior U.S. Application(s):

<u>Serial No.</u>	<u>Filing Date</u>	<u>Status: Patented, Pending, Abandoned</u>
12/390,509	02/23/2009	Pending

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I hereby revoke any previous Powers of Attorney and appoint the following attorney(s) and/or agent(s), each said individual being a member or associate of **Horizon IP Pte. Ltd.** and associated with:

Customer Number: 31366

for so long as they remain with such law offices or company with full power of substitution and revocation, to prosecute this application and to transact in connection therewith all business in the Patent and Trademark Office and before competent International Authorities; said appointment to be to the exclusion of myself and my other attorney(s).

Correspondence should be addressed to the address associated with:

Customer Number: 31366

.....

Inventor's signature: _____

Date: 03/07/2009

Full name of sole or first inventor:

Guo Wei ZHANG

Citizenship:

SG

Residence Address:

Block 426, Woodlands Street 41, #10-206,
Singapore 730426

P.O. Address:

Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Inventor's signature: 

Date: 3/07/09

Full name of second inventor:

Purakh Raj VERMA

Citizenship:

SG

Residence Address:

29 Rosewood Drive, #04-23, Casablanca,
Singapore 737291

P.O. Address:

Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/550,571

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	20 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	380
N/A	620
N/A	250
x 60 =	0.00
x 250 =	0.00
	0.00
	0.00
TOTAL	1250

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*		Minus	**	=
Independent (37 CFR 1.16(h))	*		Minus	***	=
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*		Minus	**	=
Independent (37 CFR 1.16(h))	*		Minus	***	=
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

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The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/550,571	07/16/2012	2812	1250	CSM P 2009 NAT 03 US 2	20	3

CONFIRMATION NO. 1089

16162
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

FILING RECEIPT



Date Mailed: 07/30/2012

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Guo Wei ZHANG, Singapore, SINGAPORE;
Purakh Raj VERMA, Singapore, SINGAPORE;

Assignment For Published Patent Application

GLOBALFOUNDRIES SINGAPORE PTE. LTD., Singapore, SINGAPORE

Power of Attorney: The patent practitioners associated with Customer Number 31366

Domestic Priority data as claimed by applicant

This application is a DIV of 12/500,620 07/10/2009 PAT 8222130 *
which is a CIP of 12/390,509 02/23/2009 PAT 8053319
(*)Data provided by applicant is not consistent with PTO records.

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

If Required, Foreign Filing License Granted: 07/26/2012

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/550,571**

Projected Publication Date: 11/08/2012

Non-Publication Request: No

Early Publication Request: No

Title

HIGH VOLTAGE DEVICE

Preliminary Class

438

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/550,571	07/16/2012	Guo Wei ZHANG	CSM P 2009 NAT 03 US 2	1089
16162	7590	10/26/2012	EXAMINER	
Horizon IP Pte Ltd.			HOANG, QUOC DINH	
7500A Beach Road, #04-306/308				
The Plaza			ART UNIT	PAPER NUMBER
Singapore, 199591			2892	
SINGAPORE			NOTIFICATION DATE	DELIVERY MODE
			10/26/2012	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

oa.notification@horizonip.com.sg
dexter.chin@horizonip.com.sg
info@horizonip.com.sg

Office Action Summary	Application No.	Applicant(s)	
	13/550,571	ZHANG ET AL.	
	Examiner	Art Unit	
	QUOC HOANG	2892	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 16 July 2012.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 1-20 is/are pending in the application.
- 5a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 1-20 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on ____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
 Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
 Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 12) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 13) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413) |
| 2) ☐ Notice of Draftperson's Patent Drawing Review (PTO-948) | Paper No(s)/Mail Date. ____. |
| 3) ☒ Information Disclosure Statement(s) (PTO/SB/08) | 5) ☐ Notice of Informal Patent Application |
| Paper No(s)/Mail Date <u>07/12</u> . | 6) ☐ Other: ____. |

DETAILED ACTION

Claim Rejections - 35 USC § 102

1. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

2. Claims 1, 4, and 7-12 are rejected under 35 U.S.C. 102(b) as being anticipated by Negoro et al (US Pat 6,979,864).

Regarding claim 1, Negoro teaches a device comprising:

s substrate (1, Fig. 1) prepared with an active device region, wherein the active device region includes a gate (11g, Fig. 1);

a doped channel well (23, col. 15, lines 8-37, Fig. 1) disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge defines an effective channel length of the device, wherein the effective channel length is self-aligned to the gate; and

a doped drift well (24, col. 15, lines 38-59, Fig. 1) adjacent to a second edge of the gate.

Regarding claim 4, Negoro teaches wherein the first edge of the gate (11g) corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device (drain 11d, col. 15, lines 8-59, Fig. 1).

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Regarding claims 7-12, Negoro teaches the channel length (col. 15, lines 19-21, and Fig. 1).

The limitation must distinguish from the prior art in terms of structure rather than function, *In re Schreiber*, 128 F.3d 1473, 1477-78, 44 USPQ2d 1429, 1431-32 (Fed. Cir. 1997); See also *In re Swinehart*, 439 F.2d 210, 212-13, 169 USPQ 226, 228-29 (CCPA 1971). Claims directed to apparatus must be distinguished from the prior art in terms of structure rather than function. *In re Danly*, 263 F. 2d 844, 847, 120 USPQ 528, 531 (CCPA 1959). "Apparatus claims cover what a device is, not what a device does." *Hewlett-Packard Co. v. Bausch & Lomb Inc.*, 909 F. 2d 1464, 1469, 15 USPQ2d 1525, 1528 (Fed. Cir. 1990).

Claim Rejections - 35 USC § 103

3. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

4. Claim 2 is rejected under 35 U.S.C. 103(a) as being unpatentable over Negoro et al (US Pat 6,979,864) in view of Matocha et al (US Pat 7,829,402).

Negoro teaches the channel length (col. 15, lines 19-21, and Fig. 1), but does not teach wherein the effective channel length is less than about 0.4 μm .

Matocha teaches wherein the effective channel length is less than about 0.4 μm (col. 4, lines 40-55, and Fig. 2). Since Matocha and Negoro are all from the same field

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of endeavor, the purpose disclosed by Matocha would have been recognized in the pertinent art of Negoro. It would have been obvious to a person of ordinary skill in the art at the time of the invention was made to form the effective channel length in order to enable the device with better performance and greater reliability as taught by Matocha , col. 1, lines 20-23.

5. Claims 3, 5, 6, 13, and 15-20 are rejected under 35 U.S.C. 103(a) as being unpatentable over Negoro et al (US Pat 6,979,864) in view of Chang (US Pat 7,935,992).

Regarding claims 3, 5, 6, 13, 15-17, and 20, Negoro teaches a device comprising:

s substrate (1, Fig. 1) prepared with an active device region, wherein the active device region includes a gate (11g, Fig. 1);

a doped channel well (23, col. 15, lines 8-37, Fig. 1) disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge defines an effective channel length of the device, wherein the effective channel length is self-aligned to the gate;

a doped drift well (24, col. 15, lines 38-59, Fig. 1) adjacent to a second edge of the gate; a doped deep well (21, col. 14, lines 59-60) in the substrate of the active device region encompassing the doped drift well; and first and second diffusion regions (11s 11d, Fig. 1) adjacent to the gate.

Negoro teaches the doped drift well (24, col. 15, lines 38-59, Fig. 1), but does not teach a drift isolation region in the doped drift well.

Chang teaches a drift isolation region (15) in the doped drift well (7L) (col. 3, lines 44-67, and Fig. 2A). Since Chang and Negoro are all from the same field of endeavor, the purpose disclosed by Chang would have been recognized in the pertinent art of Negoro. It would have been obvious to a person of ordinary skill in the art at the time of the invention was made to form the drift isolation region in order to define the device active region as taught by Chang, col. 1, lines 45-55.

Regarding claims 18 and 19, Negoro teaches the channel length (col. 15, lines 19-21, and Fig. 1).

The limitation must distinguish from the prior art in terms of structure rather than function, *In re Schreiber*, 128 F.3d 1473, 1477-78, 44 USPQ2d 1429, 1431-32 (Fed. Cir. 1997); See also *In re Swinehart*, 439 F.2d 210, 212-13, 169 USPQ 226, 228-29 (CCPA 1971). Claims directed to apparatus must be distinguished from the prior art in terms of structure rather than function. *In re Danly*, 263 F. 2d 844, 847, 120 USPQ 528, 531 (CCPA 1959). "Apparatus claims cover what a device is, not what a device does." *Hewlett-Packard Co. v. Bausch & Lomb Inc.*, 909 F. 2d 1464, 1469, 15 USPQ2d 1525, 1528 (Fed. Cir. 1990).

Conclusion

6. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Quoc Hoang whose telephone number is (571) 272-

Art Unit: 2892

1780. The examiner can normally be reached on Monday-Friday from 8.00 AM to 5.00 PM.

If attempt to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thao Le can be reached on (571) 272-1708. The fax phone numbers of the organization where this application or proceeding is assigned are (571) 273-8300 for regular communications and (571) 273-8300 for After Final communications.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/Quoc D Hoang/

Primary Examiner, Art Unit 2892

Notice of References Cited	Application/Control No. 13/550,571	Applicant(s)/Patent Under Reexamination ZHANG ET AL.	
	Examiner QUOC HOANG	Art Unit 2892	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-6,979,864	12-2005	Negoro et al.	257/335
*	B	US-7,935,992	05-2011	Chang, Hoon	257/288
*	C	US-7,829,402	11-2010	Matocha et al.	438/197
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Receipt date: 07/16/2012

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

13550571 - GAI: 2892

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Guowei ZHANG	
	Art Unit		
	Examiner Name	Not Assigned Yet	
	Attorney Docket Number	CSM P 2009 NAT 03 US 2	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	7776700		2010-08-17	Yang et al.	
	2	5589414		1996-12-31	Wann et al.	
	3	6759675		2004-07-06	Csutak et al.	
	4	7180132		2007-02-20	Cai et al.	
	5	6894350		2005-05-17	Shimizu et al.	
	6	6979864		2005-12-27	Negoro et al.	
If you wish to add additional U.S. Patent citation information please click the Add button.						Add
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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear

Receipt date: 07/16/2012 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13550571 - GAU: 2892	
	Filing Date			
	First Named Inventor	Guowei ZHANG		
	Art Unit			
	Examiner Name	Not Assigned Yet		
	Attorney Docket Number	CSM P 2009 NAT 03 US 2		

	1	20090059111		2009-03-05	JANG	
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	1	LEONARD RUBIN ET AL., Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, 01 July 2003, pages 119-125, Volume 46 Issue 7, Solid State Technology, USA.						☐

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE			
Examiner Signature	/Quoc Hoang/	Date Considered	10/20/2012

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Receipt date: 07/16/2012 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13550571 - GAU: 2892
	Filing Date		
	First Named Inventor	Guowei ZHANG	
	Art Unit		
	Examiner Name	Not Assigned Yet	
	Attorney Docket Number	CSM P 2009 NAT 03 US 2	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/dexter chin/	Date (YYYY-MM-DD)	2012-07-16
Name/Print	Dexter CHIN	Registration Number	38,842

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /Q.H./

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1	("7829402").PN.	USPAT; USOCR	OR	OFF	2012/10/20 13:22
L2	1	("7935992").PN.	USPAT; USOCR	OR	OFF	2012/10/20 13:36
S1	0	"13550571"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/19 18:52
S2	70	((guo near wei near zhang) (purakh near raj)).in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/19 18:55
S3	4	S2 and (substrate and active and gate and channel near well and edge and drift).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/19 18:57
S4	3	S2 and (channel near well and edge and drift and channel near length).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/19 19:12
S5	6	((("7776700") or ("5589414") or ("6759675") or ("7180132") or ("6894350") or ("6979864"))).PN.	USPAT; USOCR	OR	OFF	2012/10/19 19:13
S6	1	("20090059111").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/10/19 19:19
S7	33	overlap\$3 with channel near well with gate and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/19 19:21
S8	27	channel near well and gate near electrode and short near channel near length and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/19 20:13
S9	2	(drift and channel) near well and gate near electrode and channel near length and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/19 20:17
S10	899	drift and channel near (region well) and gate near electrode and channel near length and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/20 10:27

S11	42	isolation with drift and channel near (region well) and gate near electrode and channel near length and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/10/20 11:01
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10/ 20/ 2012 2:13:19 PM

<i>Index of Claims</i> 	Application/Control No. 13550571	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	10/20/2012								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
	6	✓								
	7	✓								
	8	✓								
	9	✓								
	10	✓								
	11	✓								
	12	✓								
	13	✓								
	14	✓								
	15	✓								
	16	✓								
	17	✓								
	18	✓								
	19	✓								
	20	✓								

Search Notes 	Application/Control No. 13550571	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

SEARCHED			
Class	Subclass	Date	Examiner
257	335,346	10/20/2012	QH

SEARCH NOTES		
Search Notes	Date	Examiner
East (search history printout)	10/20/2012	QH

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

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BIB DATA SHEET

CONFIRMATION NO. 1089

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
13/550,571	07/16/2012	257	2892	CSM P 2009 NAT 03 US 2		
APPLICANTS Guo Wei ZHANG, Singapore, SINGAPORE; Purakh Raj VERMA, Singapore, SINGAPORE; ** CONTINUING DATA ***** This application is a DIV of 12/500,620 07/10/2009 PAT 8,222,130 * which is a CIP of 12/390,509 02/23/2009 PAT 8,053,319 (*)Data provided by applicant is not consistent with PTO records. ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 07/26/2012						
Foreign Priority claimed ☐ Yes ☒ No	35 USC 119(a-d) conditions met ☐ Yes ☒ No	☐ Met after Allowance QH	STATE OR COUNTRY	SHEETS DRAWINGS	TOTAL CLAIMS	INDEPENDENT CLAIMS
Verified and /QUOC DINH HOANG/ Examiner's Signature		Initials	SINGAPORE	29	20	3
ADDRESS Horizon IP Pte Ltd. 7500A Beach Road, #04-306/308 The Plaza Singapore, 199591 SINGAPORE						
TITLE HIGH VOLTAGE DEVICE						
FILING FEE RECEIVED 1250	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees			
			☐ 1.16 Fees (Filing)			
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/550,571	07/16/2012	Guo Wei ZHANG	CSM P 2009 NAT 03 US 2

CONFIRMATION NO. 1089

PUBLICATION NOTICE



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16162
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

Title:HIGH VOLTAGE DEVICE

Publication No.US-2012-0280318-A1

Publication Date:11/08/2012

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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Amendments to the Claims

The listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (currently amended) A device comprising:

[[s]]a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate defines a an effective channel length of the device, wherein the ~~effective channel length~~ is self-aligned to the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and

a doped drift well adjacent to a second edge of the gate.

2. (currently amended) The device of claim 1 ~~wherein the effective channel length is less than about 0.4 μm~~ further comprising spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

3. (original) The device of claim 1 wherein the substrate comprises:

a drift isolation region in the doped drift well; and

a doped deep well in the substrate of the active device region encompassing the doped drift well.

4. (original) The device of claim 1 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

5. (original) The device of claim 3 wherein the drift isolation region is disposed under the gate.

6. (currently amended) The device of claim 5 wherein the first and a second ~~diffusion~~ heavily doped regions are provided adjacent to the gate, wherein:

the first ~~diffusion~~ heavily doped region is disposed in the doped channel well on a source side of the device; and

the second ~~diffusion~~ heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

7. (currently amended) The device of claim 1 wherein the ~~effective~~ channel having a channel length that is controlled by ion implantation.

8. (original) The device of claim 7 wherein the ion implantation comprises a tilt angle implant.

9. (original) The device of claim 8 wherein a tilt angle of the ion implantation is about 1-45°.

10. (original) The device of claim 7 wherein the ion implantation comprises multiple tilt angle implants which are rotated about a plane of the substrate.

11. (original) The device of claim 10 wherein a tilt angle of the implant is about 1-45°.

12. (original) The device of claim 7 wherein the ion implantation comprises a quad tilt angle implant which is rotated about a plane of the substrate.

13. (currently amended) A device comprising:

[[s]]a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

a doped channel well disposed in the substrate adjacent to a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate defines a an ~~effective-channel length~~ of the device, wherein the ~~effective-channel length~~ is self-aligned to the gate channel is displaced from a first heavily doped region adjacent to the first edge of the gate;

a doped drift well adjacent to a second edge of the gate; and

a drift isolation region in the doped drift well.

14. (currently amended) The device of claim 13 ~~wherein the effective-channel length is less than about 0.4 μm~~ further comprising a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

15. (original) The device of claim 13 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

16. (currently amended) The device of claim 13 wherein the drift isolation region is disposed under the gate.

17. (currently amended) The device of claim 13 wherein the first and a second ~~diffusion~~ heavily doped regions are provided adjacent to the gate, wherein:

the first ~~diffusion~~ heavily doped region is disposed in the doped channel well on a source side of the device; and

the second ~~diffusion~~ heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

18. (currently amended) The device of claim 13 wherein the ~~effective~~ channel having a channel length that is controlled by ion implantation.

19. (original) The device of claim 18 wherein the ion implantation comprises a tilt angle implant, multiple tilt angle implants or quad tilt angle implant.

20. (currently amended) A semiconductor device comprising:

[[s]]a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

first and second heavily doped regions adjacent to first and second edges of the gate;

a doped channel well disposed in the substrate adjacent to [[a]] the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate defines a an effective channel length of the device, wherein the effective channel length is self-aligned to the gate channel is displaced from the first heavily doped region adjacent to the first edge of the gate; and

a doped drift well adjacent to a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well; ~~and~~

~~first and second diffusion regions adjacent to the gate.~~

Remarks/Arguments

Claim Amendments

Without conceding the propriety of the rejections herein and in the interest of expediting prosecution, Applicants amend some claims herein. Applicants amend claims to clarify claimed features. Such amendments are made to expedite prosecution and more quickly identify allowable subject matter. Such amendments are merely intended to clarify the claimed features, and should not be construed as further limiting the claimed invention in response to the cited references. All amendments are fully supported by the Application and therefore do not constitute new matter.

Claim Rejections – 35 USC § 102

Claims 1, 4, and 7-12 are rejected under 35 USC § 102(b) as being anticipated by US Patent No. 6,979,864 (Negoro). Applicants respectfully disagree with this basis of rejection.

With respect to independent claim 1, Applicants respectfully submit that Negoro fails to disclose at least the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. In contrast, Negoro discloses the surface of the p-type channel well region (element 23) under the gate electrode (element 11g) is operative as a channel region. The inner edge of the n-type heavily doped region (element 11s) and the inner edge of the p-type channel well define the channel. Therefore, Negoro fails to disclose that the channel is displaced from the n-type heavily doped region. *See, e.g., Negoro*, column 15, lines 19-21; FIG. 1.

Claims 4 and 7-12 ultimately depend upon independent claim 1. As discussed above, claim 1 is allowable. It is axiomatic that any dependent claim which depends from an

allowable base claim is also allowable. Additionally, some or all of these claims may also be allowable for additional independent reasons.

Consequently, Negoro fails to disclose “each and every element” as set forth in these claims. Accordingly, Applicants ask the Examiner to withdraw the rejection of these claims.

Claim Rejections – 35 USC § 103

Claim 2 is rejected under 35 USC § 103(a) as being unpatentable over Negoro in view of US Patent No. 7,829,402 (Matocha). Applicants respectfully disagree with this basis of rejection.

Claim 2 ultimately depends on independent claim 1. As already discussed, independent claim 1 is patentable over Negoro. In particular, Negoro fails to disclose at least the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. To compensate for the defects of Negoro, the Examiner relies on Matocha. Matocha, like Negoro, fails to teach or suggest these features of the independent claim 1. This is evidenced by the fact that the Examiner only relies on Matocha to teach the channel length. Applicants therefore submit that independent claim 1, as well as its directly or indirectly dependent claims, including claim 2, are patentable over Negoro and Matocha, alone or in combination.

Claims 3, 5-6, 13, and 15-20 are rejected under 35 USC § 103(a) as being unpatentable over Negoro in view of US Patent No. 7,935,992 (Chang). Applicants respectfully disagree with this basis of rejection.

Claims 3 and 5-6 ultimately depend on independent claim 1. As already discussed, independent claim 1 is patentable over Negoro. It is axiomatic that any dependent claim which depends from an allowable base claim is also allowable. Additionally, these claims may also be allowable for additional independent reasons. Accordingly, Applicants

respectfully submit that this basis of rejection has been traversed and respectfully request its withdrawal.

With respect to independent claims 13 and 20, these claims generally recite a device. Claim 13 is representative. This claim recites a device which includes a substrate prepared with an active device region. The active device region includes a gate. The gate includes a gate electrode over a gate dielectric. The device also includes a doped channel well. The doped channel well is disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge beneath the gate define a channel of the device. The channel is displaced from a first heavily doped region adjacent to the first edge of the gate. The device also includes a doped drift well adjacent to a second edge of the gate and a drift isolation region in the doped drift well.

In the Office Action (page 5), the Examiner admits that Negoro fails to teach a drift isolation region in the doped drift well and relies on Chang to compensate for its defects. However, the addition of Chang fails to compensate for the defects of Negoro. For example, Chang also fails to teach or suggest at least the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. Therefore, the combined teaching of Negoro and Chang at most, teaches or suggests a device having a channel defined by the edge of the heavily doped (source/drain) region and the edge of the channel well. As such, one of ordinary skill in the art would not be led to forming a device having a channel which is displaced from the edge of a first heavily doped region adjacent to the first edge of the gate as recited in the present claims.

Accordingly, for at least the reasons discussed above, the asserted combination of Negoro and Chang does not teach or suggest all of the elements and features of these claims. Applicants submit that independent claims 13 and 20 as well as their directly and indirectly

Appl. No. 13/550,571
Amdt dated January 25, 2013
Reply to Office Action dated October 26, 2012

dependent claims are patentable over Negoro and Chang, alone or in combination. Accordingly, Applicants respectfully request the Examiner to withdraw the rejection of these claims.

Conclusion

It is respectfully submitted that this application is in condition for allowance and the issuance of a formal Notice of Allowance at an early date is respectfully requested.

Should the Examiner believe that a telephone conference would expedite prosecution of this application, please telephone the undersigned attorney at his number set out below.

Date: January 25, 2013

Respectfully submitted,

/dexter chin/

Dexter CHIN
Attorney for Applicants
Reg. No.: 38,842

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TRANSMITTAL FORM (to be used for all correspondence after initial filing)	Application Number	13/550,571	
	Filing Date	07/16/2012	
	First Named Inventor	Guo Wei ZHANG	
	Art Unit	2892	
	Examiner Name	HOANG, QUOC DINH	
Total Number of Pages in This Submission	12	Attorney Docket Number	CSM P 2009 NAT 03 US 2

ENCLOSURES (Check all that apply)		
☐ Fee Transmittal Form ☐ Fee Attached ☒ Amendment/Reply ☐ After Final ☐ Affidavits/declaration(s) ☐ Extension of Time Request ☐ Express Abandonment Request ☐ Information Disclosure Statement ☐ Certified Copy of Priority Document(s) ☐ Reply to Missing Parts/ Incomplete Application ☐ Reply to Missing Parts under 37 CFR 1.52 or 1.53	☐ Drawing(s) ☐ Licensing-related Papers ☐ Petition ☐ Petition to Convert to a Provisional Application ☐ Power of Attorney, Revocation Change of Correspondence Address ☐ Terminal Disclaimer ☐ Request for Refund ☐ CD, Number of CD(s) _____ ☐ Landscape Table on CD	☐ After Allowance Communication to TC ☐ Appeal Communication to Board of Appeals and Interferences ☐ Appeal Communication to TC (Appeal Notice, Brief, Reply Brief) ☐ Proprietary Information ☐ Status Letter ☐ Other Enclosure(s) (please identify below):
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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Application Number:	13550571
International Application Number:	
Confirmation Number:	1089
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Guo Wei ZHANG
Customer Number:	16162
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 2
Receipt Date:	25-JAN-2013
Filing Date:	16-JUL-2012
Time Stamp:	01:31:11
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Amendment/Req. Reconsideration-After Non-Final Reject	CSMP2009NAT03US2-ResponsetoNonFinalOA.pdf	130847 2d5ba3459f280cbb9a445a5a0183221be16014	no	10

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Information:

2	Miscellaneous Incoming Letter	CSMP2009NAT03US2-TF.pdf	294013	no	2
			45de55e876740d6b9daa01b30115650b1e b39834		
Warnings:					
Information:					
Total Files Size (in bytes):				424860	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 13/550,571		Filing Date 07/16/2012		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL		TOTAL			
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	01/25/2013	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	X \$	=		X \$62=	0
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0	X \$	=		X \$250=	0
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	0
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=		X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=		X \$	=
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/CAROL BARNES/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

16162 7590 04/15/2013
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

EXAMINER

HOANG, QUOC DINH

ART UNIT

PAPER NUMBER

2892

DATE MAILED: 04/15/2013

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/550,571

07/16/2012

Guo Wei ZHANG

CSM P 2009 NAT 03 US 2

1089

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1780	\$300	\$0	\$2080	07/15/2013

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

16162 7590 04/15/2013
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/550,571	07/16/2012	Guo Wei ZHANG	CSM P 2009 NAT 03 US 2	1089

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1780	\$300	\$0	\$2080	07/15/2013

EXAMINER	ART UNIT	CLASS-SUBCLASS
HOANG, QUOC DINH	2892	257-335000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).
☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list
(1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

NOTE: Absent a valid certification of Micro Entity Status (see form PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

☐ Applicant asserting small entity status. See 37 CFR 1.27

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

☐ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/550,571	07/16/2012	Guo Wei ZHANG	CSM P 2009 NAT 03 US 2	1089

16162 7590 04/15/2013
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

EXAMINER

HOANG, QUOC DINH

ART UNIT PAPER NUMBER

2892

DATE MAILED: 04/15/2013

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 0 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 0 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 13/550,571	Applicant(s) ZHANG ET AL.	
	Examiner QUOC HOANG	Art Unit 2892	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 01/25/2013.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Interim copies:

a) ☐ All b) ☐ Some c) ☐ None of the: Interim copies of the priority documents have been received.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.

☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____	6. ☐ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material	7. ☐ Other _____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	

/Quoc D Hoang/ Primary Examiner, Art Unit 2892	
---	--

Issue Classification 	Application/Control No. 13550571	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

CPC		
Symbol	Type	Version

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

US ORIGINAL CLASSIFICATION						INTERNATIONAL CLASSIFICATION														
CLASS		SUBCLASS				CLAIMED					NON-CLAIMED									
257		335				H	0	1	L	29 / 76 (2006.01.01)										
CROSS REFERENCE(S)																				
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)																			
257	E29.256																			

NONE		Total Claims Allowed:	
		20	
(Assistant Examiner)	(Date)		
/QUOC HOANG/ Primary Examiner.Art Unit 2892	04/07/2013	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

Issue Classification 	Application/Control No. 13550571	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

☐ Claims renumbered in the same order as presented by applicant															
☐ CPA															
☐ T.D.															
☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	17	17												
2	2	18	18												
3	3	19	19												
6	4	20	20												
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10	10														
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12	12														
13	13														
14	14														
15	15														
16	16														

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	20	
/QUOC HOANG/ Primary Examiner.Art Unit 2892	04/07/2013	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1	"13550571"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:06
L2	76	((Guo near Wei near ZHANG)(Purakh near Raj near VERMA)).in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:17
L3	0	2 and (gate and well and edge and channel and displac\$3 and heavily near doped near region).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:20
L4	0	2 and (gate and well and edge and channel and heavily near doped near region).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:21
L5	0	2 and (gate and well and edge and channel and heavily near doped).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:21
L10	3244	257/335,346.ccls. 257/E29.256.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:44
L11	9614	gate with overlap\$4 with channel and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:45
L12	201	11 and 10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2013/04/07 14:45

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L6	72	((Guo near Wei near ZHANG)(Purakh near Raj near VERMA)).in.	US-PGPUB; USPAT; UPAD	OR	OFF	2013/04/07 14:21
L7	0	6 and (gate and well and edge and channel and heavily near doped).clm.	US-PGPUB; USPAT; UPAD	OR	OFF	2013/04/07 14:21
L8	5	6 and (gate and well and edge and channel).clm.	US-PGPUB; USPAT; UPAD	OR	OFF	2013/04/07 14:21
L9	0	6 and (gate and well and edge and channel and displac\$3).clm.	US-PGPUB; USPAT; UPAD	OR	OFF	2013/04/07 14:22

4/ 7/ 2013 3:01:57 PM



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BIB DATA SHEET

CONFIRMATION NO. 1089

SERIAL NUMBER 13/550,571	FILING or 371(c) DATE 07/16/2012 RULE	CLASS 257	GROUP ART UNIT 2892	ATTORNEY DOCKET NO. CSM P 2009 NAT 03 US 2		
APPLICANTS Guo Wei ZHANG, Singapore, SINGAPORE; Purakh Raj VERMA, Singapore, SINGAPORE; ** CONTINUING DATA ***** This application is a DIV of 12/500,620 07/10/2009 PAT 8222130 * which is a CIP of 12/390,509 02/23/2009 PAT 8053319 (*)Data provided by applicant is not consistent with PTO records. ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 07/26/2012						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /QUOC DINH HOANG/ Acknowledged Examiner's Signature		☐ Met after Allowance QH Initials	STATE OR COUNTRY SINGAPORE	SHEETS DRAWINGS 29	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS Horizon IP Pte Ltd. 7500A Beach Road, #04-306/308 The Plaza Singapore, 199591 SINGAPORE						
TITLE HIGH VOLTAGE DEVICE						
FILING FEE RECEIVED 1250	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
			☐ Credit			

<i>Index of Claims</i> 	Application/Control No. 13550571	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	10/20/2012	04/07/2013						
1	1	✓	=						
2	2	✓	=						
3	3	✓	=						
6	4	✓	=						
4	5	✓	=						
5	6	✓	=						
7	7	✓	=						
8	8	✓	=						
9	9	✓	=						
10	10	✓	=						
11	11	✓	=						
12	12	✓	=						
13	13	✓	=						
14	14	✓	=						
15	15	✓	=						
16	16	✓	=						
17	17	✓	=						
18	18	✓	=						
19	19	✓	=						
20	20	✓	=						

Search Notes 	Application/Control No. 13550571	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
257	335,346	10/20/2012	QH
257	E29.256	4/7/2013	QH

SEARCH NOTES		
Search Notes	Date	Examiner
East (search history printout)	10/20/2012	QH
East (interference search history printout)	4/7/2013	QH
US_PGPub database	4/7/2013	QH

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
257	335	4/7/2013	QH

--	--

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
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INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

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16162 7590 04/15/2013
Horizon IP Pte Ltd.
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The Plaza
Singapore, 199591
SINGAPORE

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

Mei Sin TAN	(Depositor's name)
/mei sin tan/	(Signature)
07/08/2013	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/550,571	07/16/2012	Guo Wei ZHANG	CSM P 2009 NAT 03 US 2	1089

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1780	\$300	\$0	\$2080	07/15/2013

EXAMINER	ART UNIT	CLASS-SUBCLASS
HOANG, QUOC DINH	2892	257-335000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Horizon IP Pte Ltd

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

GLOBALFOUNDRIES Singapore Pte. Ltd.

Singapore

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
- ☒ Publication Fee (No small entity discount permitted)
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- ☐ A check is enclosed.
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- ☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 502388 (enclose an extra copy of this form).

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see form PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /dexter chin/

Date 07/08/2013

Typed or printed name Dexter CHIN

Registration No. 38,842

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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Mei Sin TAN	(Depositor's name)
/mei sin tan/	(Signature)
07/08/2013	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/550,571	07/16/2012	Guo Wei ZHANG	CSM P 2009 NAT 03 US 2	1089

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1780	\$300	\$0	\$2080	07/15/2013

EXAMINER	ART UNIT	CLASS-SUBCLASS
HOANG, QUOC DINH	2892	257-335000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

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1 Horizon IP Pte Ltd

2 _____

3 _____

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(A) NAME OF ASSIGNEE

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GLOBALFOUNDRIES Singapore Pte. Ltd.

Singapore

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

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- ☒ Issue Fee
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Authorized Signature _____/dexter chin/

Date _____07/08/2013

Typed or printed name _____Dexter CHIN

Registration No. _____38,842

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TRANSMITTAL FORM <i>(to be used for all correspondence after initial filing)</i>	Application Number	13/550,571	
	Filing Date	07/16/2012	
	First Named Inventor	Guo Wei ZHANG	
	Art Unit	2892	
	Examiner Name	HOANG, QUOC DINH	
Total Number of Pages in This Submission	8	Attorney Docket Number	CSM P 2009 NAT 03 US 2

ENCLOSURES (Check all that apply)		
☒ Fee Transmittal Form ☒ Fee Attached ☐ Amendment/Reply ☐ After Final ☐ Affidavits/declaration(s) ☐ Extension of Time Request ☐ Express Abandonment Request ☐ Information Disclosure Statement ☐ Certified Copy of Priority Document(s) ☐ Reply to Missing Parts/ Incomplete Application ☐ Reply to Missing Parts under 37 CFR 1.52 or 1.53	☐ Drawing(s) ☐ Licensing-related Papers ☐ Petition ☐ Petition to Convert to a Provisional Application ☐ Power of Attorney, Revocation Change of Correspondence Address ☐ Terminal Disclaimer ☐ Request for Refund ☐ CD, Number of CD(s) _____ ☐ Landscape Table on CD	☐ After Allowance Communication to TC ☐ Appeal Communication to Board of Appeals and Interferences ☐ Appeal Communication to TC (Appeal Notice, Brief, Reply Brief) ☐ Proprietary Information ☐ Status Letter ☒ Other Enclosure(s) (please identify below):
Remarks Part B - Issue Fee(s) Transmittal (1 original & 1 copy)		

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT			
Firm Name	Horizon IP Pte. Ltd.		
Signature	/dexter chin/		
Printed name	Dexter CHIN		
Date	07/08/2013	Reg. No.	38,842

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Typed or printed name	Mei Sin TAN	Date	07/08/2013

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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	13550571			
Filing Date:	16-Jul-2012			
Title of Invention:	HIGH VOLTAGE DEVICE			
First Named Inventor/Applicant Name:	Guo Wei ZHANG			
Filer:	Dexter Kam Chin			
Attorney Docket Number:	CSM P 2009 NAT 03 US 2			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl Issue Fee	1501	1	1780	1780
Publ. Fee- Early, Voluntary, or Normal	1504	1	300	300

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				2080

Electronic Acknowledgement Receipt

EFS ID:	16257991
Application Number:	13550571
International Application Number:	
Confirmation Number:	1089
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Guo Wei ZHANG
Customer Number:	16162
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 2
Receipt Date:	08-JUL-2013
Filing Date:	16-JUL-2012
Time Stamp:	22:07:49
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 2080
RAM confirmation Number	7749
Deposit Account	502388
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.20 (Post Issuance fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	PartBTransmittal-copy.pdf	2024745	no	2
			ab14826de66678e7d5606f8eb231eb8a07e1c2fc		
Warnings:					
Information:					
2	Issue Fee Payment (PTO-85B)	PartBTransmittal-org.pdf	2024745	no	2
			ab14826de66678e7d5606f8eb231eb8a07e1c2fc		
Warnings:					
Information:					
3	Miscellaneous Incoming Letter	TF.pdf	263460	no	2
			e3988bdd37d3c09e7bffc46315e04341f725d8e		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	32040	no	2
			c71521ee28fae1431029f54cab27ecd8ceb194ff		
Warnings:					
Information:					
Total Files Size (in bytes):			4344990		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/550,571	08/13/2013	8507983	CSM P 2009 NAT 03 US 2	1089

16162 7590 07/24/2013
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

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